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CM3070 FINAL PROJECT REPORT
Machine Learning And Neural Networks Project idea 2
Deep Learning Breast Cancer Detection

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1. Introduction

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1.1 Project Background and Motivation

Breast cancer is one of the most common types of cancer in women and in the world and one of the leading causes of cancer deaths in women. The prognosis is better with an earlier diagnosis because treatment can begin sooner. Mammography is the most commonly used method for screening and detecting breast cancer, but it can be a lengthy process that depends on expert radiologists to create the best results. Human reading is limited by variability, fatigue and false negatives, particularly when the abnormality is subtle, and for dense breast tissue when the soft tissue may obscure the image.

The recent past has witnessed great success for artificial intelligence, and especially deep learning, in medical image analysis. CNNs have achieved state-of-the-art results in numerous image classification benchmarks by learning to extract image features directly from raw image pixels through training of the network parameters through supervised learning. These capabilities make CNNs particularly useful in the field of mammography

imaging, where it is often difficult or cumbersome to manually extract the required features using customary image processing methods.

Developing a solution based on deep learning techniques to analyze mammography data can potentially help with breast cancer classification. Deep learning models may not necessarily replace clinical experts in their professional performance, but instead support them in the breast cancer screening procedure by improving classifications and reducing the potential of missed diagnoses. It develops, assesses and optimizes deep learning models under realistic resource constraints for the class imbalance, the used evaluation metrics and the models' generalization performance.

1.2 Problem Statement and Project Objectives

The core problem addressed in this project is the automated binary classification of mammography images into normal/benign and cancer/malignant categories. This task presents several challenges, including high image dimensionality, visual similarity between classes, and significant class imbalance in real-world screening datasets where healthy cases vastly outnumber cancer cases.

The primary objective of the project is to design and evaluate a robust deep learning pipeline capable of performing breast cancer classification from mammogram images. To achieve this, the project adopts an iterative approach that begins with simple CNN architectures and progressively incorporates more advanced techniques, such as transfer learning, threshold optimisation, fine-tuning, and data balancing strategies.

A secondary objective is to compare model behaviour across different datasets with distinct characteristics. One dataset is used to develop and stress-test the modelling pipeline under severe class imbalance conditions, while a second dataset is used to validate whether the learned design choices can achieve high metrics to a different data distribution. This two-dataset strategy allows for a more comprehensive evaluation of model robustness and avoids drawing conclusions from a single source of data.

Performance evaluation is not limited to accuracy alone. Given the medical context of the problem, particular emphasis is placed on recall, precision, F1-score, confusion matrices, and Receiver Operating Characteristic (ROC) curves. These metrics provide a more informative assessment of model performance, especially in scenarios where false negatives can have serious clinical consequences.

1.3 Project Template and Scope

This project follows the **Machine learning and neural network** template idea 2 **Deep learning breast cancer detection** idea provided by the University of London. The template focuses on applying convolutional neural networks to mammography data, specifically using the Digital Database for Screening Mammography (DDSM), and encourages exploration beyond basic CNNs through techniques such as regularisation and transfer learning.

In line with the template, the project begins by modelling the DDSM dataset using a series of progressively improved neural network architectures. Initial

models are intentionally simple to establish a baseline, after which architectural complexity and training strategies are incrementally refined. Transfer learning using pre-trained convolutional backbones is introduced to leverage features learned from large-scale image datasets, improving performance under limited computational resources.

The scope of the project is deliberately constrained to binary image classification and does not attempt lesion localisation or segmentation. The focus remains on model design, training methodology, evaluation strategy, and comparative analysis rather than clinical deployment. All experiments are conducted using supervised learning techniques, with datasets split into training, validation, and test subsets to ensure fair and unbiased evaluation.

The final outcome of the project is a comparative analysis of multiple deep learning models, supported by quantitative performance metrics and qualitative discussion. By following a structured deep learning workflow and evaluating models across two datasets and then deploying the best performing model into a web app where the users can upload a picture and get whether it's cancer or normal. the project aims to demonstrate both technical competence and critical understanding of applied machine learning in a medical imaging context.

2. Literature Review

2446 WORDS without tables

2.1 Breast Cancer Screening and the Role of Mammography

Mammography is the most widely used screening tool for breast cancer because it is accessible, relatively inexpensive, and effective for early detection. It allows radiologists to identify suspicious findings such as masses and calcifications that may indicate malignancy. However, interpretation remains difficult, particularly in women with dense breast tissue or subtle abnormalities. The growing volume of mammographic examinations also places pressure on radiologists and may affect diagnostic consistency.

These challenges have encouraged interest in computer-assisted systems that can support clinical decision-making. Wang et al. (2024) note that although mammography remains the gold standard for breast cancer screening, diagnostic performance can vary according to patient factors and image quality. This has increased interest in artificial intelligence methods that may improve the consistency and sensitivity of cancer detection. Such findings provide a strong rationale for investigating deep learning approaches within the mammography screening workflow.

2.2 Traditional Computer-Aided Diagnosis and Its Limitations

Before deep learning became widespread, computer-aided diagnosis (CAD) systems relied on hand-crafted image features combined with classical machine learning algorithms. These systems extracted descriptors such as texture, shape, or intensity from mammograms and used classifiers such as support vector machines or k-nearest neighbours. Although they demonstrated some success, their performance depended heavily on the quality of manual feature engineering and often failed to generalise across datasets.

Wang et al. (2024) argue that hand-crafted features are often insufficient for representing the subtle and high-dimensional patterns associated with breast cancer. Traditional CAD systems also require substantial domain expertise to design and tune features, reducing scalability and adaptability to new data. These limitations contributed to the shift toward deep learning methods, which can learn discriminative features directly from image data and therefore offer a more flexible alternative.

2.3 Convolutional Neural Networks in Mammography Analysis

Convolutional neural networks (CNNs) have become the dominant model family in computer vision due to their ability to learn hierarchical image representations. In medical imaging, CNNs are particularly valuable because they can automatically learn both low-level and high-level visual features without the need for manual feature extraction. Abdelrahman et al. (2021) report that CNN-based approaches generally outperform traditional CAD systems in mammography tasks.

Their review also suggests that while simple CNN architectures can be effective, deeper architectures typically achieve stronger discriminative performance. However, CNN results are strongly influenced by dataset size, class balance, and preprocessing strategy. This makes CNNs suitable as a

foundation for breast cancer detection, but also indicates that performance depends on careful dataset handling and model design.

Geras et al. (2017) proposed a multi-view CNN designed for high-resolution mammography, preserving fine image detail rather than aggressively downsampling inputs. Using a very large clinical screening dataset, the model achieved an AUC of 0.895 and demonstrated that combining multiple mammographic views improves classification performance. This study highlights the importance of preserving diagnostically relevant information in mammography analysis.

McKinney et al. (2020) evaluated a deep learning system for breast cancer screening using mammograms from the United Kingdom and the United States. Their model processed full mammography exams end-to-end and achieved higher diagnostic performance than the average radiologist, increasing AUC from 0.750 to 0.871 while reducing false positives and false negatives. These results demonstrate the clinical potential of large-scale deep learning systems.

Chang et al. (2025) extended this discussion by reporting results from the AI-STREAM prospective multicentre study. Their findings showed that integrating AI into radiologists' workflows increased cancer detection without increasing recall rates, and improved the detection of smaller, earlier-stage cancers.

While these studies collectively demonstrate the effectiveness of deep learning in mammography analysis, they differ in their evaluation contexts. For example, Geras et al. (2017) focus on architectural improvements for high-resolution image processing, whereas McKinney et al. (2020) evaluate a large-scale screening system against radiologists across international datasets. In contrast, Chang et al. (2025) provide prospective clinical evidence of AI integration into routine screening workflows. Together, these studies illustrate the progression from model development toward real-world clinical validation of AI systems.

2.4 Public Mammography Datasets and Data Challenges

Publicly available mammography datasets play a central role in developing and evaluating automated breast cancer detection systems. Among these, the Digital Database for Screening Mammography (DDSM) remains one of the most widely used benchmark datasets because of its size and accessibility (Amin et al., 2025). It has therefore become a common resource for training and comparing deep learning models.

However, public datasets also introduce important challenges. Amin et al. (2025), in a systematic review of 287 mammography studies, identify class imbalance, variations in acquisition protocols, and limited diversity in patient populations as key issues affecting model development. These problems can introduce dataset bias and restrict generalisation to different clinical settings.

Zufiria et al. (2022) provide experimental evidence of this issue by showing that models trained on screening datasets can perform substantially worse on diagnostic datasets because of differences in scanners, protocols, and image distributions. Together, these studies demonstrate that dataset characteristics strongly influence the reliability of mammography models. They also justify the need to examine dataset properties carefully and to evaluate models across more than one dataset whenever possible.

2.5 Class Imbalance and Evaluation Metrics in Medical Diagnosis

Class imbalance is one of the most important challenges in medical imaging, particularly in screening datasets where normal cases heavily outnumber abnormal ones. In breast cancer detection, this can lead to misleading performance estimates if evaluation relies mainly on accuracy. Reviews by Wang et al. (2024) and Amin et al. (2025) both note that a model can achieve high accuracy while still failing to detect cancer cases reliably.

For this reason, the literature increasingly recommends metrics such as recall, precision, F1-score, and ROC-based analysis. These metrics provide a more informative view of model behaviour in imbalanced settings, particularly

because false negatives are costly in medical diagnosis. This supports the use of recall-focused evaluation in breast cancer classification.

Gao et al. (2020) examined severe class imbalance in medical imaging and proposed a one-class classification framework to improve anomaly detection. Evaluated on several medical datasets, including DDSM, the method achieved an AUC of 0.941 and demonstrated that specialised training strategies are often needed in highly imbalanced medical tasks.

Saito and Rehmsmeier (2015) further showed that ROC curves may give overly optimistic performance estimates when negative examples dominate the dataset. In contrast, precision–recall analysis better reflects the ability of a model to identify rare positive cases. Their work suggests that metrics focusing on the minority class, such as recall and F1-score, are often more meaningful for medical diagnosis. Together, these studies show that both training strategy and evaluation design must be adapted when dealing with imbalanced mammography data.

2.6 Transfer Learning for Medical Image Classification

Transfer learning has become a widely used strategy in medical imaging because annotated medical datasets are often limited. Instead of training deep networks from scratch, researchers can adapt models pre-trained on large image datasets such as ImageNet. Mendel et al. (2019) note that pre-trained CNNs often converge faster and generalise better than randomly initialised models in medical imaging applications.

Sahu et al. (2024) similarly report that architectures such as ResNet and DenseNet perform well across different breast imaging tasks, while other studies show that fine-tuning higher layers can improve domain adaptation. This supports the exploration of multiple pre-trained architectures in mammography classification.

Kim et al. (2022), in a systematic review of 121 medical imaging studies, found that transfer learning consistently improves performance, reduces training

time, and lowers computational cost. Architectures such as ResNet, Inception, and VGG were commonly used as feature extractors, with fine-tuning applied to adapt the learned features to the target domain. These findings suggest that transfer learning itself may be more important than any single backbone architecture, and strongly justify its use in projects with limited labelled data.

2.7 Deep CNN Architectures and Performance Trends

Deeper CNN architectures generally offer greater representational power than shallow models, allowing them to learn more complex patterns in medical images. Tsochatzidis et al. (2019) found that deeper models tended to outperform simpler ones in mammography classification, especially when dataset size and regularisation were sufficient.

However, the literature also warns that deeper models are more susceptible to overfitting, particularly in medical datasets with limited training examples. As a result, regularisation, data augmentation, and careful fine-tuning are often recommended to balance complexity and generalisation. These findings support a progressive modelling strategy in which complexity is increased gradually and validated experimentally rather than assuming that deeper networks will always perform best.

2.8 State-of-the-Art Models

Recent studies on ResNet50 and DenseNet architectures for breast cancer detection highlight critical trade-offs in optimization, explainability, and generalization across diverse imaging domains (Zhang et al., 2023; Houssein et al., 2022). While ResNet50-based approaches emphasize fast, stable convergence and false-positive control, DenseNet variants leverage dense feature reuse to handle imbalanced data more effectively, though both remain

limited by dataset-specific training and weak cross-site validation (Liu et al., 2024; Talukder, 2025).

Zhang et al. (2023) combine Mask R-CNN with a ResNet50 classifier on breast MRI, demonstrating a two-stage pipeline . first localize lesions, then classify using fine-tuned ImageNet-pretrained ResNet50 over 100 epochs with Adam and moderate batch size. The model achieves 96% per-exam sensitivity at only 0.1 false positives per image, largely because Mask R-CNN filters out the majority of irrelevant regions before the classifier acts. This end-to-end pipeline is highly effective for reducing radiologist workload in screening, but its MRI-only design makes it a weak reference for pure mammography systems and introduces a dependency on segmentation quality errors at the first stage cascade directly into misclassification (Zhang et al., 2023).

Houssein et al. (2022) adopt a very different strategy: they keep ResNet50 purely as a classifier on mammography-like datasets (MIAS and CBIS-DDSM) and instead optimize the training pipeline using an improved marine predators algorithm to tune hyperparameters .This automation yields 98.2% accuracy, 97.5% sensitivity, and 98.7% specificity on CBIS-DDSM, with an F1-score of 0.98. Compared to Zhang et al.'s MRI-based system, Houssein et al. achieve higher raw accuracy on mammographic data and better balance between precision and recall, but they lack the explicit false-positive filtering component ; instead, they rely on aggressive data augmentation and focal loss to manage class imbalance. This makes Houssein et al. a stronger benchmark for understanding how to tune ResNet50 in mammography, though their small MIAS subset still introduces risk of overfitting and limits real-world generalization (Houssein et al., 2022).

Switching to DenseNet, Liu et al. (2024) benchmark DenseNet169 on a histopathology and mammography-inspired setup, using SMOTE

oversampling to rebalance malignant versus benign examples. Their dense architecture captures multi-scale features more efficiently than ResNet50, which leads to 92.5% accuracy and 91% F1-score despite slower training roughly three times the wall-time of the ResNet50-based methods. In contrast with Houssein et al.'s automated hyperparameter search, Liu et al. emphasize data-level balancing and dense connectivity rather than optimizer tuning. This makes their approach more robust to minority-class under-representation, but the slower training and higher computational cost reduce its practicality for real-time screening unless resources are abundant (Liu et al., 2024).

Finally, Talukder (2025) pushes DenseNet201 further by adding explicit explainable AI (XAI) via reconstruction and fine-tuning on DDSM/CBIS-DDSM, achieving 97.8% accuracy and 96.5% AUC. The model explicitly highlights suspicious regions in its outputs, providing clinicians with visual feedback that is absent in both Zhang et al.'s MRI pipeline and Houssein et al.'s highly optimized but opaque ResNet50. In this respect, Talukder's work represents a clear advancement in interpretability over the DenseNet benchmarking of Liu et al. However, it also inherits DenseNet's drawbacks: slower training and higher sensitivity to class imbalance without additional SMOTE-like adjustments. The main clinical trade-off is that better explainability comes at the cost of slightly higher false negatives in dense-breast regions, where calcifications and subtle lesions remain challenging even for state-of-the-art architectures (Talukder, 2025)

Table 1. Comparison of selected state-of-the-art deep learning architectures for breast cancer detection in mammography and related medical imaging datasets

Paper (Year)	Architecture & Dataset	Key Technique	Strength Compared to Other Studies	Main Limitation
Zhang et al. (2023)	ResNet50 + Mask R-CNN (MRI dataset)	Segmentation-then-classification pipeline	Reduces false positives and provides a clear detection workflow for lesion localisation	Limited to MRI data; performance depends on segmentation quality
Houssein et al. (2022)	ResNet50 (MIAS and CBIS-DDSM datasets)	Marine-Predators-based hyperparameter optimisation	Achieved very high classification accuracy and F1-score on mammography datasets	Risk of overfitting due to small dataset size and lack of explainability methods
Liu et al. (2024)	DenseNet169 (histopathology and mammography-inspired datasets)	SMOTE-based class balancing	Improved feature reuse through dense connections and strong performance on	Increased computational cost and longer training time

			imbalanced datasets	
Talukder (2025)	DenseNet201 + XAI (DDSM and CBIS-DDSM datasets)	Reconstruction-based explainable AI approach	Provides interpretable visual explanations that can support clinical decision making	Higher false negative rates in dense breast tissue and slower training

In summary, Houssein et al. (2022) outperform Zhang et al. (2023) in pure accuracy on mammographic data but lack explicit false-positive reduction and visualization, while Liu et al. (2024) and Talukder (2025) demonstrate that DenseNet-based architectures can surpass ResNet50 in both F1-score and interpretability especially when combined with SMOTE and XAI. These gaps highlight the need for integrated pipelines that combine ResNet50's speed and robustness with DenseNet's feature-reuse strength and explicit explainability, which is precisely the direction taken by the proposed system in this project.

2.9 Validation and Generalisation Across Datasets

A recurring limitation in the literature is the lack of external validation on independent datasets. Amin et al. (2025) observe that many studies report strong results on a single dataset but do not test whether these results hold

under different data conditions. This raises concerns about dataset bias and real-world robustness.

Garrucho et al. (2022) investigated domain generalisation in mammography mass detection using a large multi-centre study. They found that models trained on data from one centre often experienced substantial performance degradation when evaluated on images from different institutions. Variations in equipment, acquisition protocols, and patient populations all contributed to this decline. Although domain generalisation techniques can reduce these effects, the study concluded that multi-dataset evaluation remains essential for assessing real robustness.

This contrasts with many earlier studies that report strong performance using only one dataset. Garrucho et al. (2022) therefore reinforce the importance of evaluating deep learning systems across multiple datasets when possible, especially for projects that aim to make claims about robustness rather than only dataset-specific accuracy.

2.10 Summary of Literature and Research Gap

The reviewed literature shows that deep learning, particularly CNN-based methods, has become the dominant approach for automated breast cancer detection in mammography. Studies such as Geras et al. (2017), McKinney et al. (2020), and Chang et al. (2025) demonstrate that deep learning systems can achieve strong diagnostic performance and increasingly support clinical workflows. At the same time, the literature identifies major challenges including class imbalance, dataset bias, limited validation across datasets, and the risk of relying too heavily on accuracy as a performance metric.

Research on class imbalance and evaluation metrics shows that recall, precision, and F1-score are often more informative than accuracy in highly imbalanced medical tasks (Saito and Rehmsmeier, 2015). Transfer learning studies further suggest that pre-trained architectures such as ResNet and DenseNet can improve medical image classification while reducing training

cost (Kim et al., 2022). Recent state-of-the-art studies also show that architecture choice, balancing strategy, and interpretability methods influence performance, but many systems remain dataset-specific and insufficiently validated.

Therefore, a clear research gap remains in the systematic evaluation of deep learning pipelines that address class imbalance while also assessing robustness across datasets. This project addresses that gap by implementing a progressive model development pipeline, applying optimisation techniques such as threshold adjustment, fine-tuning, and oversampling, and evaluating performance across two mammography datasets with different data characteristics.

3. Design

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3.1 Project Overview

This project aims to develop an end-to-end deep learning pipeline for breast cancer detection from mammography images, with a particular focus on handling class imbalance and limited annotated data. The design follows an incremental approach, beginning with a baseline CNN and progressively introducing transfer learning, threshold optimisation, fine-tuning, and data balancing so that each improvement can be evaluated separately. Because false negatives are especially harmful in medical diagnosis, model evaluation prioritises recall and F1-score over overall accuracy. The final outcome is a reproducible and interpretable deep learning model capable of classifying mammograms as normal or cancerous.

3.2 Domain, Users, and Intended Use

Domain

The domain of this project is medical image analysis, specifically breast cancer screening using X-ray mammography. Mammography is a widely used diagnostic tool, but its interpretation is challenging and subject to inter-observer variability, particularly in early-stage cancer detection.

Intended Users

The system is designed for:

- academic researchers exploring computer-aided diagnosis (CAD),

- medical imaging researchers studying deep learning applications,
- and as a proof-of-concept decision support tool rather than a clinical replacement for radiologists .

Despite the importance of radiologists as the main user of breast cancer detection systems in mammography screenings, our system is designed as a decision-support tool, rather than an independent breast cancer diagnostic system. Radiologists increasingly rely on computer-helped detection (CAD) and artificial intelligence systems to screen images, as established by prior research. For example, Masud et al. (2019) found that CAD solutions improved the sensitivity of cancer detection and decreased reading time. Radiologists prefer double reading because of high recall rates and costs, however (Masud et al. , 2019). Recent data suggest that AI-based systems are becoming more widely adopted for clinical use. According to a survey of Society of Breast Imaging (SBI) members conducted by Nenow et al (2026), more than 50% of radiologists are currently using AI-based tools as part of their practice, mainly due to improved workflow and improved cancer detection performance. Remaining challenges include the rate of false positives, the integration into clinical workflows, and the trustworthiness of the predictions. The aim of this project was to ease research into the use of deep learning for mammography analysis rather than to make the final diagnosis, which is the role of the clinician.

Intended Use

The intended use of the system is to:

- analyse mammography images,
- output a binary classification (normal vs cancer),

- and provide confidence-aware predictions that prioritise cancer recall.

3.3 System Workflow and Architecture

The overall system workflow follows a standard deep learning pipeline, adapted to the specific requirements of medical imaging such shown in **Figure 1**

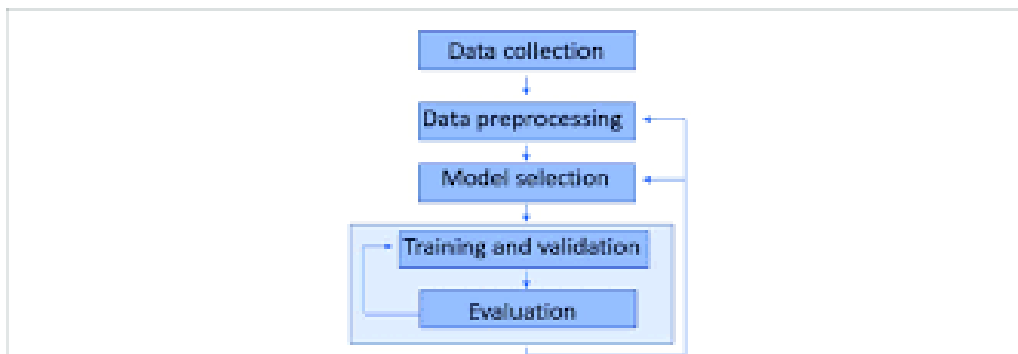


Figure 1 overall system work flow

The workflow consists of the following stages:

1. Data ingestion and preprocessing, where raw mammography images are loaded and prepared for model input through resizing, normalisation, and dataset-specific preprocessing steps.
2. Dataset splitting into training, validation, and test sets, ensuring that evaluation is performed on unseen data and that no data leakage occurs between subsets.
3. Model training using convolutional neural network architectures and transfer learning, where baseline CNN

models are first developed and progressively enhanced using pre-trained deep learning architectures.

4. Model evaluation using clinically relevant performance metrics, including confusion matrices, recall, F1-score, and ROC-AUC, with particular emphasis on reducing false negative predictions.
5. Final model selection based on balanced performance, taking into account both overall classification ability and medical relevance rather than accuracy alone.
6. Model deployment and system integration, where the selected final model is incorporated into a lightweight deployment layer designed to enable end-to-end interaction with the system. This deployment component allows a user to upload a mammography image and then produces a binary classification output (normal or cancer). The deployment is designed for demonstration and validation purposes, completing the system workflow from raw image input to interpretable prediction without modifying the trained model or evaluation strategy.

This workflow follows a typical deep learning experimentation pipeline used in medical image analysis, where each stage is designed to ensure reproducible and unbiased model evaluation. The pipeline begins with careful data preparation and dataset splitting to prevent data leakage, followed by incremental model development where increasingly complex architectures are introduced. Evaluation is then performed using clinically relevant metrics that emphasise the reduction of false negatives. Finally, the selected model is deployed within a simple user interface to demonstrate how the trained model can be integrated into an end-to-end system capable of processing raw images and generating predictions. This modular design allows each stage to be analysed and justified independently.

3.4 Dataset Design and Data Management

Two datasets are used in this project, each serving a distinct design purpose.

Dataset 1: DDSM

The Digital Database for Screening Mammography (DDSM) is used as the primary development dataset. This dataset contains grayscale mammography images labelled as either normal or cancerous.

Key characteristics:

- Large-scale dataset with severe class imbalance,
- Cancer cases represent a minority of samples,
- Images require resizing and normalisation before training.

The primary purpose of this dataset is to:

- stress-test the models under realistic imbalance conditions,
- explore threshold optimisation strategies,
- and evaluate the impact of fine-tuning and data balancing.

DDSM cancer sample shown

Figure 2.DDSM CANCER SAMPLE



Dataset 2: Breast Mammography Images with Masses

The second dataset consists of breast mammography images stored in folder-based structure, labelled as benign (0) or malignant (1).

Key characteristics:

- Images stored as individual files,
- More balanced class distribution compared to DDSM,
- Used as a validation dataset to assess if we can achieve higher metrics on balanced dataset .

This dataset is intentionally designed into the project to verify if we can achieve higher metrics and better performance on a balanced dataset.

DATASET2 CANCER SAMPLE SHOWN

Figure 3.DATASET2 CANCER SAMPLE

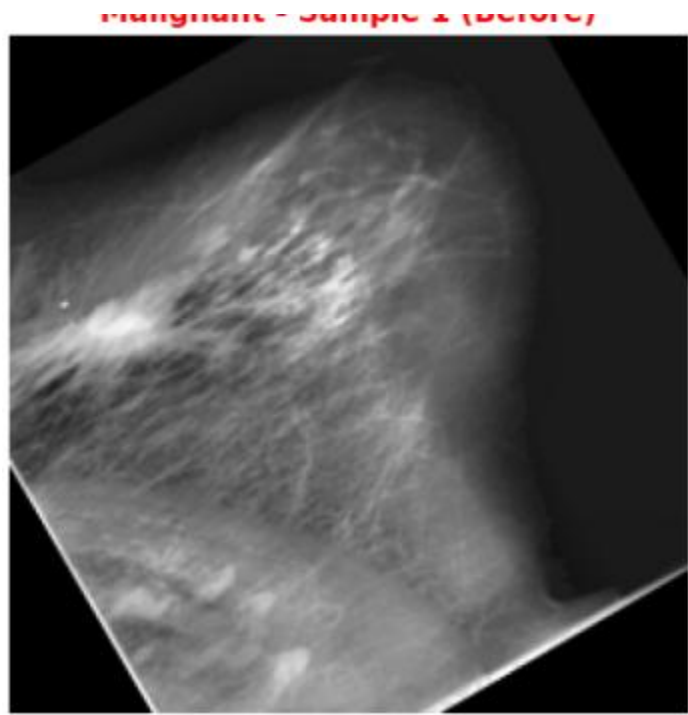


Table 2 shows the key charactrstics of each dataset with all its numbers and its role in the project

Table 2. Dataset Characteristics

Dataset Name	Source	Task	Number of Samples	Class Labels	Class Distribution	Role in Project
DDSM	Kaggle	Binary Classification	55,890 training samples	Normal (0), Cancer (1)	~86% Normal, ~14% Cancer	Primary development dataset for model design, optimisation, and imbalance handling
Breast Mammography Images with Masses	kaggle	Binary Classification	24,600 images	Benign (0), Malignant (1)	~10.9k Benign, ~13.7k Malignant	Model development to test if we can achieve higher metrics on balanced dataset

Data Leakage Prevention

A critical design decision is the strict separation of training, validation, and test data. Oversampling and augmentation are applied only to the training set, ensuring that no duplicated samples appear in validation or test data.

This prevents artificially inflated performance metrics and maintains the integrity of evaluation results.

3.5 Data Preprocessing Strategy

Preprocessing strategies differ between datasets while maintaining a consistent modelling philosophy.

Preprocessing for Dataset 1 (DDSM)

- Images are resized to a fixed resolution,
- Normalised for neural network input,
- Converted to three-channel format where required by pre-trained models.

Preprocessing for Dataset 2

- Images are converted to grayscale,
- Contrast and brightness correction (CBC) is applied,
- Images are resized to 224×224 pixels,
- Data augmentation is applied to the training set only.

Table 3 shows again more characteristics and what preprocessing happens to each dataset and why the preprocessing differs

Table 3. Characteristics and preprocessing of Datasets

Aspect	Dataset 1 (DDSM)	Dataset 2 (Masses – Kaggle)
Storage format	TfRecord files	Folder-based image files
Image type	Grayscale mammograms	Grayscale mammograms
Image resolution	Resized during preprocessing	Resized to 224×224
Class balance	Severely imbalanced	Moderately balanced
Preprocessing	Resizing, normalisation, channel conversion	CBC filtering, resizing, normalisation
Data augmentation	Not applied	Applied to training set only
Purpose	Model development and stress-testing	Model development to see if we can achieve higher metrics on balanced data

The design deliberately avoids excessive preprocessing to preserve clinically relevant image features.

3.6 Model Development Strategy

The model development strategy follows a progressive experimental approach, where models of increasing complexity are evaluated sequentially. This design allows the contribution of each technique to be analysed independently. Starting with a simple CNN baseline provides a reference point for evaluating improvements introduced by transfer learning, threshold optimisation, fine-tuning, and class balancing techniques. Such incremental experimentation is common in deep learning research because it enables systematic comparison of architectural and training modifications.

Baseline CNN Models

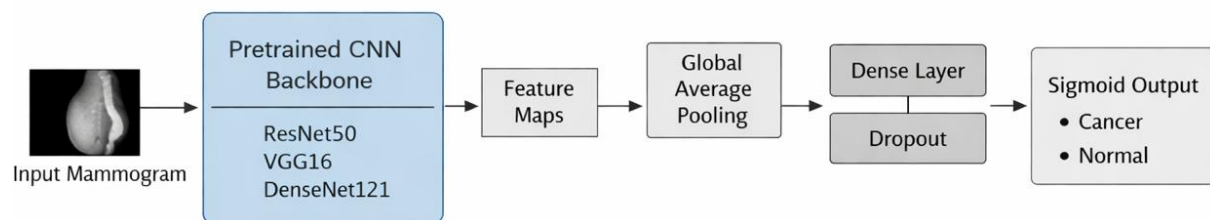
Initial models consist of simple CNN architectures trained from scratch. These models establish baseline performance and highlight the limitations of naive approaches under class imbalance.

Transfer Learning Models

Transfer learning models such as VGG16, DenseNet121, and ResNet50 are introduced using pre-trained ImageNet weights. Transfer learning is widely used in medical image classification because labelled medical datasets are often limited in size. Pre-trained convolutional networks trained on large image datasets can learn general visual features that transfer effectively to medical imaging tasks (Kim et al., 2022). By using transfer learning, the models can achieve improved performance while requiring significantly less

training data and computational time compared to training networks from scratch

Figure 4. Transfer learning architecture used for mammography classification



Threshold Optimisation

In this project, the classification decision threshold is not fixed at the conventional value of 0.5. Instead, the threshold is explicitly optimized based on the F1-score to achieve a better balance between precision and recall. This design choice is particularly important in medical classification tasks where datasets are often imbalanced and false negatives can have serious consequences. Previous research has shown that adjusting the decision threshold can significantly improve performance for probabilistic classifiers when dealing with imbalanced data (Lipton et al., 2015). By optimising the

threshold according to the F1-score, the model increases sensitivity for detecting cancer cases while maintaining reasonable precision, aligning the evaluation strategy with the clinical priority of minimizing missed cancer diagnoses

Fine-Tuning Strategy

For the best-performing transfer learning model, selected layers are unfrozen and retrained using a reduced learning rate. This allows the model to adapt high-level features to mammography-specific patterns without overfitting.

Class Imbalance Handling

To address the severe class imbalance present in the DDSM dataset, minority class oversampling is applied exclusively to the training data. Oversampling increases the representation of cancer samples during training, reducing the likelihood that the classifier becomes biased toward the majority benign class. Previous research has shown that class imbalance can significantly degrade the performance of machine learning models by causing classifiers to favour the dominant class, particularly in medical imaging tasks where abnormal cases are relatively rare (Gao et al., 2020). Oversampling strategies have been proposed as an effective approach for mitigating this issue by improving the model's ability to learn features associated with minority classes without altering the test distribution (Walsh and Tardy, 2022). By increasing the proportion of cancer samples during training while maintaining

untouched validation and test sets, the model is encouraged to detect malignant patterns more reliably while preserving unbiased evaluation. This approach is particularly important in medical imaging applications where false negatives carry significant clinical risk

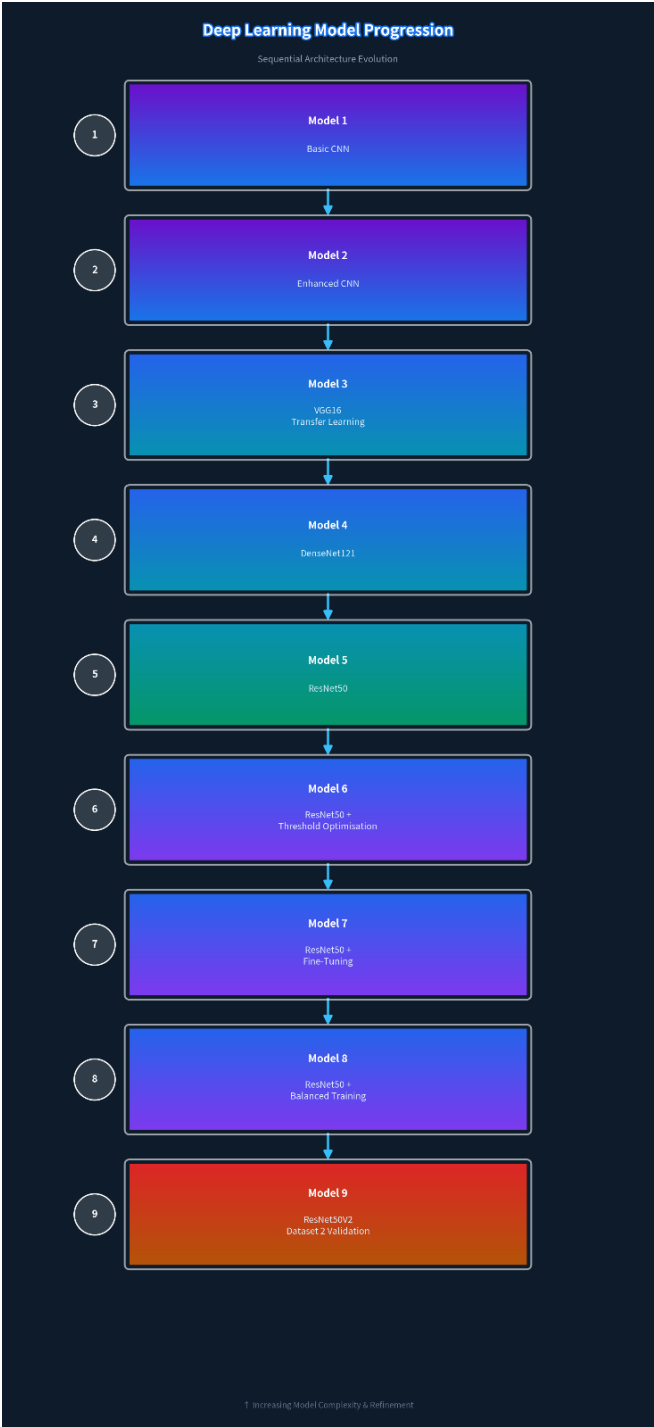
Table 4 will show why each model was chosen and what's the difference between each model in its architecture and this shows methodological experimentation and proves that the models weren't chosen randomly.

Table 4. Model Development plan

Model ID	Architecture	Key Modification	Purpose
Model 1	Basic CNN	Simple convolutional layers	Establish baseline performance
Model 2	Enhanced CNN	Deeper architecture	Improve feature extraction
Model 3	VGG16	Frozen backbone	Evaluate transfer learning
Model 4	DenseNet121	Frozen backbone	Compare deep architectures
Model 5	ResNet50	Frozen backbone	Identify best-performing transfer model
Model 6	ResNet50	Threshold optimisation	Improve recall and F1-score
Model 7	ResNet50	Fine-tuning	Adapt features to mammography data
Model 8	ResNet50	Fine-tuning + balanced training	Address severe class imbalance
Model 9	ResNet50V2	Frozen backbone	Trained on different balanced dataset to make sure that we can

			achieve high metrics on a balanced dataset
--	--	--	--

Figure 5. progressive model developement strategy used to evaluate increasly complex architectures



3.7 Technologies and Tools Used

The experiment was built around the standard deep learning experiment tools and reproducible research workflows. Google Colab was the primary computer system that was selected since it was a cloud service offering a built-in Graphics Processing Unit (GPU) to be able to execute a deep neural network without requiring special local hardware. The various convolutional neural network architectures were built using TensorFlow and Keras libraries as they provide great compatibility with transfer learning and experimentation with different deep learning methods. Precision, recall, F1-score, confusion matrices, and many other metrics were computed through the scikit-learn library for classification model evaluation and analysis. We stored the datasets, trained models, and results of our experiments in Google Drive so that all of the experiments were reproducible and the models could be retrieved for evaluation and deployment.

3.8 Work Plan and Milestones

Table 5. Milestones of the project

Week	Task	Description
Week 1 - 2	Dataset Acquisition and Exploration	Download DDSM dataset, inspect data quality, verify labels, and perform initial preprocessing tests.
Week 3 - 4	Data Preprocessing	Apply reshaping, resizing, normalization, and 3-channel conversion. Prepare data splits for training/validation/testing.
Week 5 - 7	Model Design and Implementation	Build and code the Basic CNN, Enhanced CNN, and implement pretrained architectures (ResNet50, DenseNet121, VGG16).
Week 8 - 9	Training and Hyperparameter Tuning	Train models, optimize parameters (batch size, learning rate, epochs), and log performance metrics.
Week 10	Evaluation and Comparative Analysis	Compute all metrics, generate ROC and Confusion Matrix plots, and identify the best-performing model.
Week 11 - 12	Streamlit GUI Development	Integrate the chosen model into Streamlit interface for live predictions. Add visualization outputs.
Week 12-15	Testing, Validation, and Documentation	Conduct final testing, write project documentation, and prepare report submission.

3.9 Evaluation Strategy

Evaluation focuses on:

- precision, recall, and F1-score,
- confusion matrices,
- ROC–AUC curves.

Evaluation focuses on clinically relevant metrics including precision, recall, F1-score, confusion matrices, and ROC–AUC curves. In medical diagnosis tasks such as breast cancer detection, evaluation metrics must reflect the clinical risks associated with prediction errors rather than relying solely on overall accuracy. Previous research has shown that accuracy can be misleading when applied to imbalanced medical datasets, where the majority class dominates the data distribution (Amin et al., 2025). Consequently, this project prioritises recall and F1-score for the cancer class to ensure that malignant cases are detected reliably.

Recall is particularly important in medical screening systems because false negatives represent missed cancer cases that may delay diagnosis and treatment. Studies in mammography analysis emphasise the importance of sensitivity-oriented evaluation in order to minimise such risks (Wang et al., 2024). In addition, F1-score is used to provide a balanced assessment of precision and recall under class imbalance conditions.

Confusion matrices are included to provide a transparent breakdown of prediction outcomes, allowing explicit examination of false negatives and false positives. Finally, ROC–AUC is reported to evaluate the model’s overall discriminative ability across different classification thresholds. However, as previous research has demonstrated that ROC curves can sometimes overestimate performance in highly imbalanced datasets, precision–recall related metrics remain particularly informative for medical classification tasks (Saito and Rehmsmeier, 2015). **Table 6** describes them in complete details with its description and reason for use

Table 6. Evaluation Metrics

Metric	Description	Reason for Use
Accuracy	Overall correct predictions	Reported but not relied upon due to imbalance
Precision	Correct positive predictions	Indicates reliability of cancer predictions
Recall	True positive rate	Critical to reduce false negatives
F1-score	Balance of precision and recall	Suitable for imbalanced datasets
Confusion matrix	Breakdown of predictions	Explicitly shows false negatives
ROC–AUC	Discriminative performance	Threshold-independent evaluation

3.10 Risks and Limitations

Potential risks include:

- dataset bias,
- limited annotation quality,
- lack of clinical validation.

These limitations are acknowledged explicitly, and results are interpreted within an academic research context rather than a clinical deployment setting.

4.Implementation

2127 word without tables

The implementation of this project translates the designed deep learning pipeline into a working system capable of training, evaluating, and deploying breast cancer classification models using mammography images. The implementation was carried out incrementally, with each stage building upon the previous one, allowing weaknesses to be identified and addressed through targeted improvements.

All development and experimentation were conducted using Google Colab with GPU acceleration. This environment was selected to support the computational requirements of deep convolutional neural networks while

remaining accessible and reproducible. Trained models and intermediate results were saved to Google Drive to ensure persistence across sessions and to allow later reuse during deployment.

4.1 Dataset 1: DDSM Implementation

The first dataset used in the project was the Digital Database for Screening Mammography (DDSM). This dataset is provided in TFRecord format and contains pre-labelled mammography images. The dataset exhibits a strong class imbalance, with cancer cases forming a small minority of the total samples, making it suitable for stress-testing model behaviour under realistic screening conditions.

Data Ingestion and Preprocessing

The implementation begins by reading the TFRecord files using TensorFlow's data pipeline utilities. Each record is parsed to extract the encoded image and associated label. The image data are decoded from raw byte format and reshaped into a two-dimensional grayscale image. Since several deep learning architectures expect colour images, the grayscale image is subsequently converted into a three-channel representation by duplicating the single channel across three channels.

Images are resized to a fixed resolution to ensure consistent input dimensions across all models. Normalisation is applied to scale pixel values into a suitable numerical range for neural network training. These preprocessing steps are applied consistently across training, validation, and test sets to ensure comparability. **Figure 6** shows the code of the preprocessing.

The dataset is then split into training, validation, and test subsets. Care is taken to ensure that these splits are mutually exclusive, preventing any form of data leakage. The training set is used for model learning, the validation set

for hyperparameter tuning and early stopping, and the test set for final performance evaluation.

Figure 6. CODE SNIPPET OF PREPROCESSING

We preprocess the data first by extracting the image and the label from the tfrecord

Then we preprocess it by reshaping it and resizing it (100,100) and making it 3 dimensions

```
images=[]
labels=[]
feature_dictionary = {
    'label': tf.io.FixedLenFeature([], tf.int64),
    'label_normal': tf.io.FixedLenFeature([], tf.int64),
    'image': tf.io.FixedLenFeature([], tf.string)
}

def _parse_function(example, feature_dictionary=feature_dictionary):
    parsed_example = tf.io.parse_example(example, feature_dictionary)
    return parsed_example

def read_data(filename):
    full_dataset = tf.data.TFRecordDataset(filename,num_parallel_reads=tf.data.experimental.AUTOTUNE)
    full_dataset = full_dataset.shuffle(buffer_size=31000)
    full_dataset = full_dataset.cache()
    print("Size of Training Dataset: ", len(list(full_dataset)))

    feature_dictionary = {
        'label': tf.io.FixedLenFeature([], tf.int64),
        'label_normal': tf.io.FixedLenFeature([], tf.int64),
        'image': tf.io.FixedLenFeature([], tf.string)
    }

    full_dataset = full_dataset.map(_parse_function, num_parallel_calls=tf.data.experimental.AUTOTUNE)
    print(full_dataset)
    for image_features in full_dataset:
        image = image_features['image'].numpy()
        image = tf.io.decode_raw(image_features['image'], tf.uint8)
        image = tf.reshape(image, [299, 299])
        image=image.numpy()
        image=cv2.resize(image,(100,100))
        image=cv2.merge([image,image,image])
        images.append(image)
        labels.append(image_features['label_normal'].numpy())

    for file in train_tfrecords:
        read_data(file)

print(len(images))
print(len(labels))
```

Baseline and Enhanced CNN Models

The first implemented model is a basic convolutional neural network trained from scratch. This model consists of a small number of convolutional and pooling layers followed by fully connected layers. Early stopping is enabled to prevent overfitting by monitoring validation loss during training. **Figure 7** shows the architecture of the basic cnn baseline model.

Figure 7.BASIC CNN ARCHITECTURE CODE

```

import tensorflow as tf
from tensorflow.keras import layers, models, regularizers

def create_cnn(input_shape, num_classes):
    model = models.Sequential([
        layers.Conv2D(32, (3, 3), activation='relu', input_shape=input_shape),
        layers.MaxPooling2D((2, 2)),

        layers.Conv2D(64, (3, 3), activation='relu'),
        layers.MaxPooling2D((2, 2)),

        layers.Conv2D(128, (3, 3), activation='relu'),
        layers.Flatten(),

        layers.Dense(128, activation='relu'),
        layers.Dropout(0.3),
        layers.Dense(1, activation='sigmoid')
    ])

    model.compile(optimizer='adam',
                  loss='binary_crossentropy',
                  metrics=['accuracy'])
    return model

input_shape = (100, 100, 3)

with tf.device('/GPU:0'):
    model = create_cnn(input_shape=input_shape, num_classes=1)
    model.summary()

```

While the baseline CNN achieves reasonable overall accuracy, analysis of the confusion matrix reveals that it performs poorly on the minority cancer class. This behaviour highlights the limitations of simple architectures when applied to highly imbalanced medical datasets.

To address this, an enhanced CNN model is implemented. This version increases architectural depth by adding additional convolutional layers and parameters, allowing the model to learn more complex visual features.

Although this improves performance compared to the baseline, the imbalance issue remains evident, motivating the transition to transfer learning. **Figure 8** shows the enhanced architecture of the enhanced cnn.

Figure 8. ARCHITECTURE CODE OF THE ENHANCED CNN

```
import tensorflow as tf
from tensorflow.keras import layers, models, regularizers

def create_enhanced_cnn(input_shape, num_classes=1):
    model = models.Sequential([
        layers.Conv2D(128, (3,3), activation='relu', padding='same', input_shape=input_shape),
        layers.MaxPooling2D((2,2)),

        layers.Conv2D(128, (3,3), activation='relu', padding='same'),
        layers.MaxPooling2D((2,2)),
        layers.Dropout(0.2),

        layers.Conv2D(256, (3,3), activation='relu', padding='same'),
        layers.MaxPooling2D((2,2)),
        layers.Dropout(0.2),

        layers.Conv2D(256, (3,3), activation='relu', padding='same'),
        layers.MaxPooling2D((2,2)),
        layers.Dropout(0.2),

        layers.Conv2D(512, (3,3), activation='relu', padding='same'),
        layers.MaxPooling2D((2,2)),
        layers.Dropout(0.3),

        layers.Conv2D(512, (3,3), activation='relu', padding='same'),
        layers.MaxPooling2D((2,2)),
        layers.Dropout(0.3),

        layers.Flatten(),
        layers.Dense(256, activation='relu'),
        layers.Dropout(0.3),
        layers.Dense(1, activation='sigmoid')
    ])

    model.compile(
        optimizer=tf.keras.optimizers.Adam(learning_rate=1e-4),
        loss='binary_crossentropy',
        metrics=['accuracy', tf.keras.metrics.AUC(name='AUC')]
    )

    return model
```

Transfer Learning Models

Three transfer learning models were utilized (VGG16, DenseNet121, and ResNet50), using pre-trained weights based on large visual datasets and adapted for binary mammography classification applications.

When imported, the convolutional backbones are frozen, meaning their learned feature representations are kept intact while only the added classification layers are iteratively adjusted. This allows to reduce computational cost and the risk of overfitting. The best performing model, ResNet50, was chosen as a baseline model to further improve upon. **FIGURE 9** shows the code of ResNet50 and what was explained above.

Figure 9. ResNet50 CODE

```

>
base_model = ResNet50(
    weights='imagenet',
    include_top=False,
    input_shape=(60, 60, 3)
)

for layer in base_model.layers:
    layer.trainable = False

x = base_model.output
x = GlobalAveragePooling2D()(x)
x = Dense(256, activation='relu')(x)
x = Dropout(0.4)(x)
output = Dense(1, activation='sigmoid')(x)

model_resnet = Model(inputs=base_model.input, outputs=output)

model_resnet.compile(
    optimizer=Adam(learning_rate=1e-4),
    loss='binary_crossentropy',
    metrics=['accuracy', tf.keras.metrics.AUC(name='auc')]
)

model_resnet.summary()

```

Threshold Optimisation and Fine-Tuning

Instead of a fixed threshold, threshold optimisation is then performed. In this process, the model is applied to the validation set as described above and the threshold which maximizes the F1-score is used on the predictions. This step makes it so that the decision boundary can shift towards recall without causing an important rise in false positives. **Figure 10** shows how the best threshold calculated.

Figure 10.CODE SHOWS THE CALCULATION OF THE BEST THRESHOLD

```

fpr, tpr, thresholds = roc_curve(y_test, pred_probs)
f1_scores = []
for th in thresholds:
    preds = (pred_probs > th).astype(int)
    f1_scores.append(f1_score(y_test, preds))

best_f1_threshold = thresholds[np.argmax(f1_scores)]

print("Best Threshold (F1-score):", best_f1_threshold)

```

```
Best Threshold (F1-score): 0.13916637
```

Upon completion of threshold optimization, the ResNet50 model is fine-tuned: the upper layers are unfrozen, and retrained using a lower learning rate. This allows for pre-trained feature extractors to be adapted with mammography specific features whilst still maintaining a stable model during training. **Figure 11** shows how the ResNet model layers was unfreezed to train on our data.

Figure 11.CODE SNIPPET OF FINE TUNING THE RESNET50 MODEL

```

# Unfreeze last 30 layers only (safe tuning)
for layer in model_resnet.layers[-30:]:
    layer.trainable = True

model_resnet.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=1e-5),
    loss='binary_crossentropy',
    metrics=['accuracy', tf.keras.metrics.AUC(name='AUC')]
)

model_resnet.summary()

```

Handling Class Imbalance

Despite improvements from fine-tuning and threshold optimisation, class imbalance remains a key challenge. To address this, minority class oversampling is implemented exclusively on the training set. Cancer samples are repeated to create a more balanced class distribution during training, while validation and test sets remain untouched to preserve unbiased evaluation. **Figure 12** shows how minority oversampling was carried out.

Figure 12. Code Snippet of minority repitition

```

We will oversample the train data only as we don't want repitition in test or validation

>
idx_label0 = np.where(y_train == 0)[0] # majority
idx_label1 = np.where(y_train == 1)[0] # minority

n_majority= len(idx_label0)
n_minority = len(idx_label1)

repeat_factor = n_majority // n_minority + 1
idx_label1_oversample = np.tile(idx_label1, repeat_factor)[:n_majority]

# Combine with all majority indices
idx_balanced = np.concatenate([idx_label0, idx_label1_oversample])
np.random.shuffle(idx_balanced)

X_train_balanced = X_train[idx_balanced]
y_train_balanced = y_train[idx_balanced]

print("Balanced dataset shape:", X_train_balanced.shape)
print("Balanced label counts:", np.unique(y_train_balanced, return_counts=True))

.. Balanced dataset shape: (58224, 100, 100, 3)
   Balanced label counts: (array([0, 1]), array([29112, 29112]))

```

The fine-tuned ResNet50 model is retrained using this balanced training data, and threshold optimisation is applied again. This final configuration produces the best balance between recall and precision for the cancer class and is selected as the final model for Dataset 1. **Figure 13** shows the classification

report of our best model with the confusion matrix and all the model results will be shown in the evaluation section.

Figure 13. FINE-TUNED BALANCED RESNET MODEL RESULTS

Classification Report with Optimized Threshold:				
	precision	recall	f1-score	support
0	0.97	0.98	0.97	7272
1	0.84	0.78	0.81	1111
accuracy			0.95	8383
macro avg	0.90	0.88	0.89	8383
weighted avg	0.95	0.95	0.95	8383
Confusion Matrix:				
[[7106 166]				
[242 869]]				

4.2 Dataset 2: Independent Validation Implementation

To examine the impact of class balance on achievable performance metrics, a second dataset consisting of breast mammography images with masses is used. This dataset has a more balanced class distribution and is stored as individual image files organised in directories. It is used as a complementary experiment to analyse how dataset characteristics influence model performance.

Data Preparation and Preprocessing

Images are loaded from disk, converted to grayscale, and resized to a fixed resolution. Contrast and brightness correction (CBC) filtering is applied to enhance visual features relevant to mass detection. Unlike Dataset 1, data augmentation techniques such as rotation, shifting, and zooming are applied to the training set to improve robustness. **Figure 14** shows how CBC and augmentation code was carried out .

Figure 14.CBC AND AUGMENTATION CODE

```

import numpy as np
import cv2
from PIL import Image
from tensorflow.keras.preprocessing.image import ImageDataGenerator

def apply_CBC(image, alpha=1.0, beta=0):
    """Apply CBC filter (contrast/brightness)"""
    if isinstance(image, str):
        image = cv2.imread(image, cv2.IMREAD_GRAYSCALE)
        if image is None:
            raise ValueError(f"Could not load image from path: {image}")

    if not isinstance(image, np.ndarray):
        image = np.array(image)

    if image.dtype != np.uint8:
        if image.max() <= 1.0:
            image = (image * 255).astype(np.uint8)
        else:
            image = image.astype(np.uint8)

    if len(image.shape) == 3:
        image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

    img_cbc = cv2.convertScaleAbs(image, alpha=alpha, beta=beta)
    img_cbc = cv2.resize(img_cbc, (224, 224))
    return img_cbc

print("Applying CBC filter to training data...")
X_train_filtered = np.array([apply_CBC(img, alpha=1.0, beta=0) for img in X_train], dtype='float32')
print(f"Shape after CBC filter: {X_train_filtered.shape}")

X_train_filtered = X_train_filtered.reshape(X_train_filtered.shape[0], 224, 224, 1)
print(f"Shape after reshape: {X_train_filtered.shape}")

print("\nLoading validation data...")
X_val_array = []
for img_path in X_val:
    if isinstance(img_path, str):
        img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
    else:
        img = img_path
    img = cv2.resize(img, (224, 224))
    X_val_array.append(img)
X_val_array = np.array(X_val_array, dtype='float32').reshape(-1, 224, 224, 1)
print(f"X_val shape: {X_val_array.shape}")

print("\nLoading test data...")
X_test_array = []
for img_path in X_test:
    if isinstance(img_path, str):
        img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
    else:
        img = img_path
    img = cv2.resize(img, (224, 224))
    X_test_array.append(img)
X_test_array = np.array(X_test_array, dtype='float32').reshape(-1, 224, 224, 1)
print(f"X_test shape: {X_test_array.shape}")

train_datagen = ImageDataGenerator(
    rotation_range=15,
    width_shift_range=0.1,
    height_shift_range=0.1,
    zoom_range=0.1,
    horizontal_flip=True,
    fill_mode='nearest',
    rescale=1./255
)

val_test_datagen = ImageDataGenerator(rescale=1./255)

train_generator = train_datagen.flow(X_train_filtered, y_train, batch_size=32, shuffle=True)
val_generator = val_test_datagen.flow(X_val_array, y_val, batch_size=32, shuffle=False)
test_generator = val_test_datagen.flow(X_test_array, y_test, batch_size=32, shuffle=False)

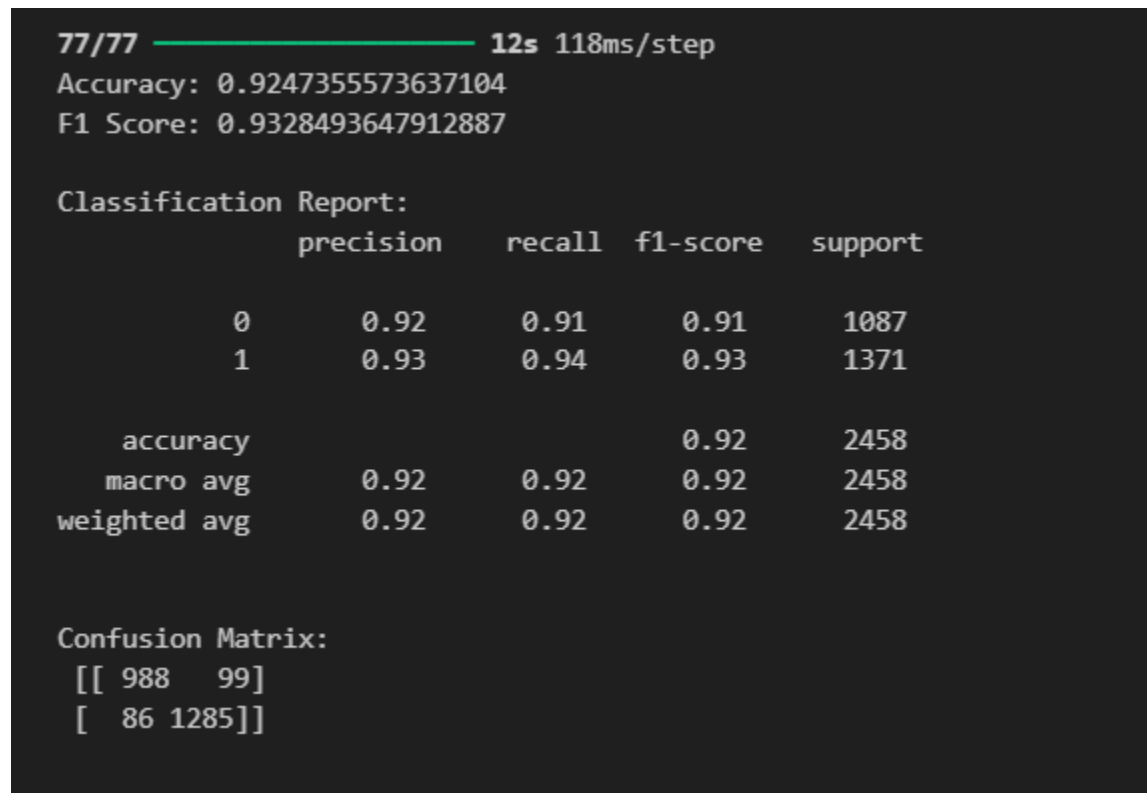
```

The dataset is split into training, validation, and test sets using stratified sampling to preserve class proportions. This ensures fair evaluation and prevents bias during model assessment.

Model Training and Evaluation

A ResNet50V2 model with a frozen backbone is trained on this dataset using the Adam optimiser, binary cross-entropy loss, and early stopping based on validation loss. The results demonstrate that higher and more balanced performance metrics can be achieved when training on a dataset with improved class balance, supporting the project's analysis of imbalance effects rather than indicating architectural superiority. **Figure 15** shows the full classification report of resnet50v2 model and shows that the results on this dataset achieved better performance.

Figure 15. RESNET50V2 classification report



4.3 Deployment Implementation

The best performing model of Dataset 1 is encapsulated in a lightweight deployment system(streamlit). The operator loads the model and can submit a mammography picture for inference. To ensure consistent results, the

uploaded images are preprocessed using the same operations as applied during model training as shown in **Figure 16**.

Figure 16.DEPLOYMENT CODE

```
def load_model():
    model = tf.keras.models.load_model("finetuned_balanced_resnet.h5")
    return model

model = load_model()

def preprocess_image(uploaded_file):
    image = Image.open(uploaded_file).convert('L')
    image = image.resize((100, 100))
    image = np.array(image)
    image = cv2.merge([image, image, image])
    image = image.astype('float32')
    image = np.expand_dims(image, axis=0)

    return image

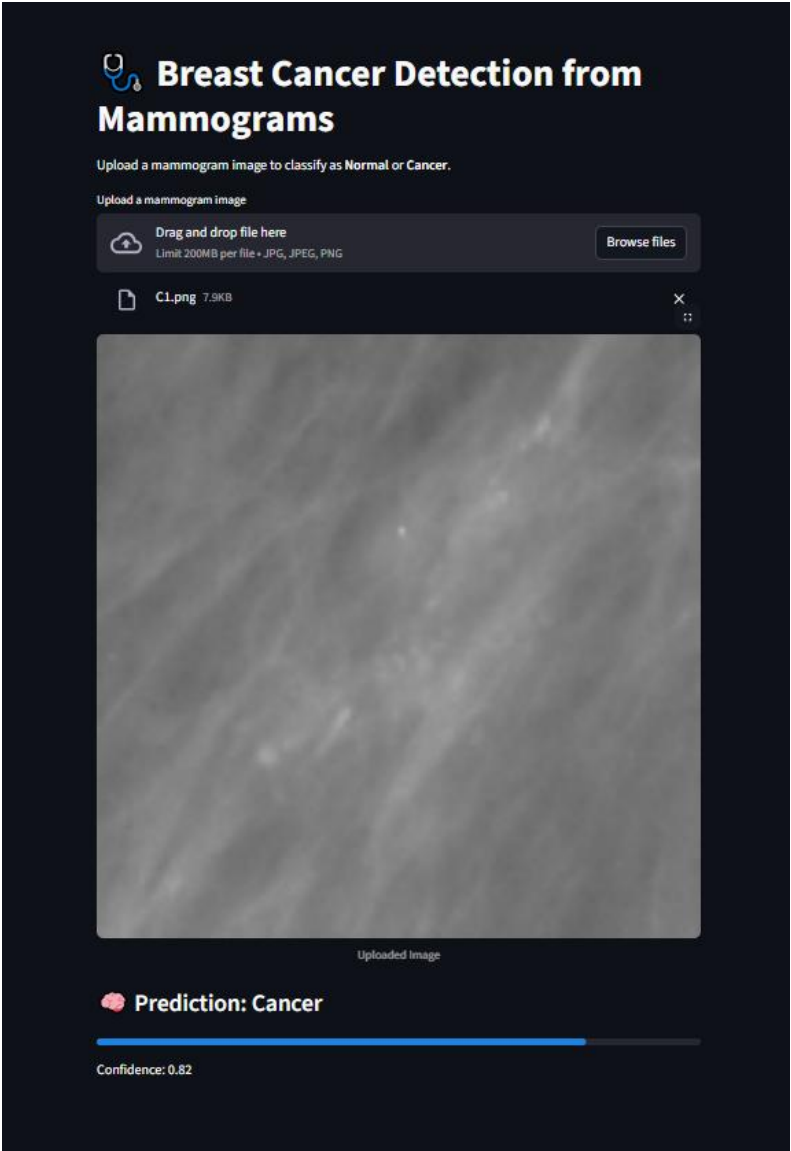
st.title("🔍 Breast Cancer Detection from Mammograms")
st.write("Upload a mammogram image to classify as Normal or Cancer.")

uploaded_file = st.file_uploader("Upload a mammogram image", type=["jpg", "jpeg", "png"])

if uploaded_file is not None:
    image = Image.open(uploaded_file)
    st.image(image, caption="Uploaded Image", use_container_width=True)
    processed_img = preprocess_image(uploaded_file)
    prediction = model.predict(processed_img)[0][0]
    label = "Cancer" if prediction > 0.20042022 else "Normal"
    confidence = prediction if prediction > 0.20042022 else 1 - prediction
    st.markdown(f"### 🟡 Prediction: {label}")
    st.progress(float(confidence))
    st.write(f"Confidence: {confidence:.2f}")
```

The output of the model is a probability and so the optimized threshold found during the evaluation stage can now be used to determine the class of the image. This allows the model to be used end-to-end, the result being returned to the user as a classification. **Figure 17** shows the UI and how the probability is shown.

Figure 17.STREAMLIT UI



The deployment portion is for demonstration and validation purposes only, and does not impact the trained model or its evaluation.

4.4 Methodology and Techniques Used

This section explains the key techniques applied during the implementation phase and how they were used to address challenges specific to medical image classification, particularly class imbalance and decision sensitivity.

CNN-Based Image Classification

Convolutional neural networks (CNNs) form the core of the implemented system due to their effectiveness in learning hierarchical visual representations from image data. In this project, CNNs are used to automatically extract discriminative features from mammography images, eliminating the need for manual feature engineering.

Baseline CNN models were first implemented to establish reference performance and evaluate how simple architectures perform on mammography data. These models highlighted the limitations of shallow architectures when applied to highly imbalanced medical datasets.

Transfer Learning Strategy

Transfer learning was used to improve model accuracy and training efficiency. Pre-trained models VGG16, DenseNet121, ResNet50, and ResNet50V2 were selected due to their strong performance in image classification tasks.

Initially, the convolutional backbones were frozen and only the added classification layers were trained. This approach leverages general-purpose visual representations while reducing overfitting and computational cost.

Among the evaluated architectures, ResNet50 consistently achieved the best performance and was therefore selected for further optimisation.

Threshold Optimisation

In standard binary classification, a probability threshold of 0.5 is commonly used to convert predicted probabilities into class labels. However, in class-imbalanced medical datasets such as breast cancer screening, the cost of false negatives is particularly high.

Threshold optimisation was therefore applied by evaluating model predictions on the validation dataset across thresholds ranging from 0 to 1. Precision, recall, and F1-score were computed for each threshold, and the threshold that maximised the F1-score was selected as the optimal operating point. This adjustment improved recall for the cancer class while reducing false negative predictions and ensured consistent behaviour during training, testing, and deployment.

Fine-Tuning of Pre-Trained Models

The selected transfer learning model (ResNet) was further fine-tuned on the mammography dataset. During the initial transfer learning stage, the convolutional base was frozen to preserve learned representations and stabilise training.

Because mammography images differ significantly from natural images, the upper layers of the network were later unfrozen and retrained with a lower

learning rate. This allowed the model to gradually adapt high-level features to mammography data while maintaining training stability.

Minority Class Oversampling

Class imbalance is a common challenge in medical imaging datasets, where abnormal cases are typically underrepresented. In the DDSM dataset, cancer samples represent a small minority compared to normal samples, which can bias models toward the majority class.

To mitigate this issue, minority class oversampling was applied during training by repeating cancer samples within the training set. Oversampling was applied exclusively to the training data, while validation and test sets remained unchanged to preserve unbiased evaluation.

By exposing the model to a more balanced class distribution during training, oversampling improved the model's ability to learn features associated with cancer cases. Combined with threshold optimisation and fine-tuning, this contributed to improved recall for the cancer class.

Contrast and Brightness Correction (CBC) Filtering

Contrast and brightness correction (CBC) filtering was applied to the second dataset during preprocessing to enhance clinically relevant structures in mammographic images. Mammography images often have low contrast, making it difficult to distinguish masses from surrounding tissue.

CBC filtering adjusts pixel intensity distributions using contrast and brightness parameters, improving the visibility of foreground structures while preserving spatial information. This preprocessing step helps CNN models better distinguish important features within the images.

Because the second dataset was stored differently from the DDSM dataset, CBC filtering provided a suitable preprocessing approach tailored to its visual characteristics.

Data Augmentation Strategy

To improve generalisation and reduce overfitting, data augmentation was applied to the second dataset. Data augmentation increases the diversity of training data by applying random transformations to existing images.

Transformations included rotations, width and height shifts, zoom operations, and horizontal flips. These augmentations simulate variations in image acquisition while preserving the underlying class labels. Augmentation was applied only to the training dataset, while validation and test sets remained unchanged to ensure unbiased evaluation.

This targeted approach reflects dataset-specific preprocessing decisions, where the DDSM dataset was primarily used to analyse class imbalance, while the second dataset was augmented to improve robustness and overall performance.

Summarization of methods

To summarise the key techniques applied during the implementation phase and their specific roles within the system, Table 7 provides an overview of the main methods used and the challenges they were designed to address.

Table 7. Methods summarization

Technique	Applied In	Purpose	Problem Addressed
CNN-Based Classification	Dataset 1 & Dataset 2	Automatically learn hierarchical features from mammography images	Manual feature engineering not suitable for medical images
Transfer Learning (VGG16, DenseNet121, ResNet50, ResNet50V2)	Dataset 1 & Dataset 2	Leverage pre-trained visual features to improve performance and training efficiency	Limited labelled medical imaging data
Threshold Optimisation	Dataset 1 (DDSM)	Adjust decision boundary based on F1-score	High false negative rate due to class imbalance
Fine-Tuning	Dataset 1 (ResNet50)	Adapt pre-trained features to mammography domain	Domain mismatch between natural images and medical images
Minority Class Oversampling	Dataset 1 (Training set only)	Increase exposure to cancer samples during training	Severe class imbalance in DDSM dataset

Contrast and Brightness Correction (CBC)	Dataset 2	Enhance visibility of clinically relevant structures	Low contrast and varying image quality
Data Augmentation	Dataset 2 (Training set only)	Improve robustness and generalisation	Overfitting and limited dataset variability

4.5 Evaluation of the Implementation

The implementation successfully achieves its intended objectives.

Progressive optimisation steps lead to significant improvements in recall and F1-score for the cancer class. The complementary experiment using a more balanced dataset confirms that class balance plays a major role in achievable performance metrics. Complete evaluation analysis will be carried out in the evaluation section.

The deployment interface demonstrates that the system operates end-to-end, from image input to interpretable output.

4.6 Limitations and Future Improvements

Training was constrained by hardware availability, limiting extensive experimentation on the severely imbalanced dataset. Dataset labels may contain noise, and no clinical validation was performed. Future work could explore larger multi-institutional datasets, alternative imbalance handling strategies, and expert-guided evaluation.

5. Evaluation

2460 words without tables

This section presents a comprehensive evaluation of the project outcomes. The evaluation focuses on assessing model performance across multiple architectures and optimisation stages, analysing the impact of class imbalance handling, and examining how results vary under different dataset conditions. In addition, the deployed user interface is evaluated to demonstrate end-to-end system usability. The evaluation is aligned with the project's primary objective of achieving high classification performance, while reducing false negative predictions and handling class imbalance as a key technical constraint.

The evaluation builds directly upon the design philosophy and implementation choices outlined in earlier sections and is informed by evaluation practices discussed in the literature.

5.1 Evaluation Strategy and Objectives

The evaluation strategy was designed to assess both the technical performance of the implemented models and the overall effectiveness of the

system as a complete solution. The primary objective of the project is to achieve high-performing breast cancer classification models. Within this objective, reducing false negative predictions is prioritised due to their critical implications in medical screening contexts, while class imbalance is treated as a constraint that directly influences achievable performance.

This evaluation strategy directly reflects the design decisions presented in the Design section. In particular, the emphasis on recall, F1-score, and confusion matrix analysis follows the design requirement to prioritise medical relevance over raw accuracy. The staged evaluation of progressively complex models mirrors the modular pipeline proposed during system design, ensuring that each design choice is empirically validated.

To support this evaluation, multiple performance metrics are employed. Confusion matrices are used to analyse prediction distributions and identify false negative and false positive cases. Recall and F1-score for the positive class are prioritised to reflect diagnostic sensitivity, while ROC curves and ROC–AUC values are used to assess discriminative ability across different decision thresholds. This choice of metrics is consistent with practices highlighted in the literature review, which emphasise the limitations of accuracy-based evaluation for imbalanced medical datasets.

In medical image classification tasks, the selection of evaluation metrics must reflect the clinical consequences of prediction errors. In breast cancer screening, false negative predictions are particularly critical because missed cancer cases may delay diagnosis and treatment. For this reason, recall (also referred to as sensitivity) is widely considered one of the most important metrics when evaluating computer-aided diagnostic systems. Previous research has emphasised that accuracy alone can be misleading in imbalanced medical datasets, as models may achieve high overall accuracy

while failing to detect minority disease cases. Studies on deep learning in mammography therefore recommend complementing accuracy with recall, F1-score, confusion matrix analysis, and ROC–AUC to provide a more clinically meaningful evaluation of model performance (Wang et al., 2024).

Evaluation is conducted progressively across different model architectures and optimisation stages using the DDSM dataset, which exhibits severe class imbalance and therefore provides a realistic stress test. A second, more balanced dataset is used to examine achievable performance under improved data conditions. Together, these evaluations provide a comprehensive assessment of the system.

5.2 Dataset 1 (DDSM): Model Performance Progression

This subsection evaluates whether the implementation choices described in the Implementation chapter translated into measurable performance improvements when applied to the DDSM dataset. As outlined in the system design, the DDSM dataset serves as the primary benchmark due to its severe class imbalance and clinical relevance.

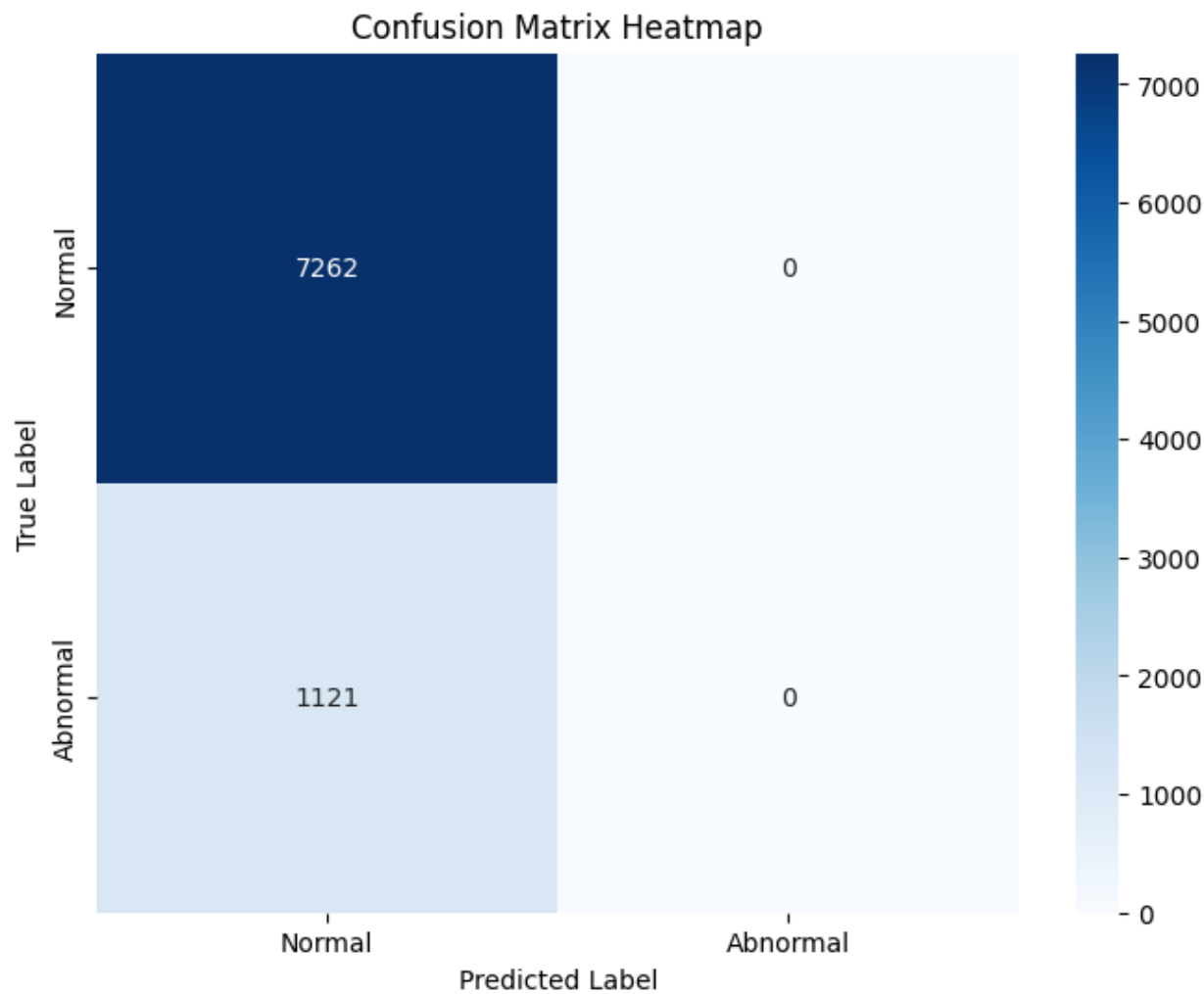
Table 8 presents a comparative summary of the performance achieved by all evaluated models on the DDSM dataset, illustrating how classification performance evolves as architectural complexity and optimisation techniques are progressively applied.

Table 8.DDSM MODELS RESULTS

Model	Accuracy	Precision (Cancer)	Recall (Cancer)	F1-score (Cancer)	ROC – AUC
Basic CNN	0.87	0.00	0.00	0.00	0.70
Enhanced CNN	0.89	0.93	0.21	0.35	0.93
VGG16 (Frozen)	0.90	0.71	0.48	0.57	0.91
DenseNet121 (Frozen)	0.91	0.76	0.50	0.60	0.93
ResNet50 (Frozen)	0.94	0.79	0.70	0.74	0.97
ResNet50 + Threshold Optimisation	0.94	0.77	0.74	0.75	0.97
FINETUNED RESNET50+THRESHOLD	0.94	0.75	0.78	0.77	0.97
Fine-tuned ResNet50 + Balanced Trainin	0.95	0.84	0.78	0.81	0.98

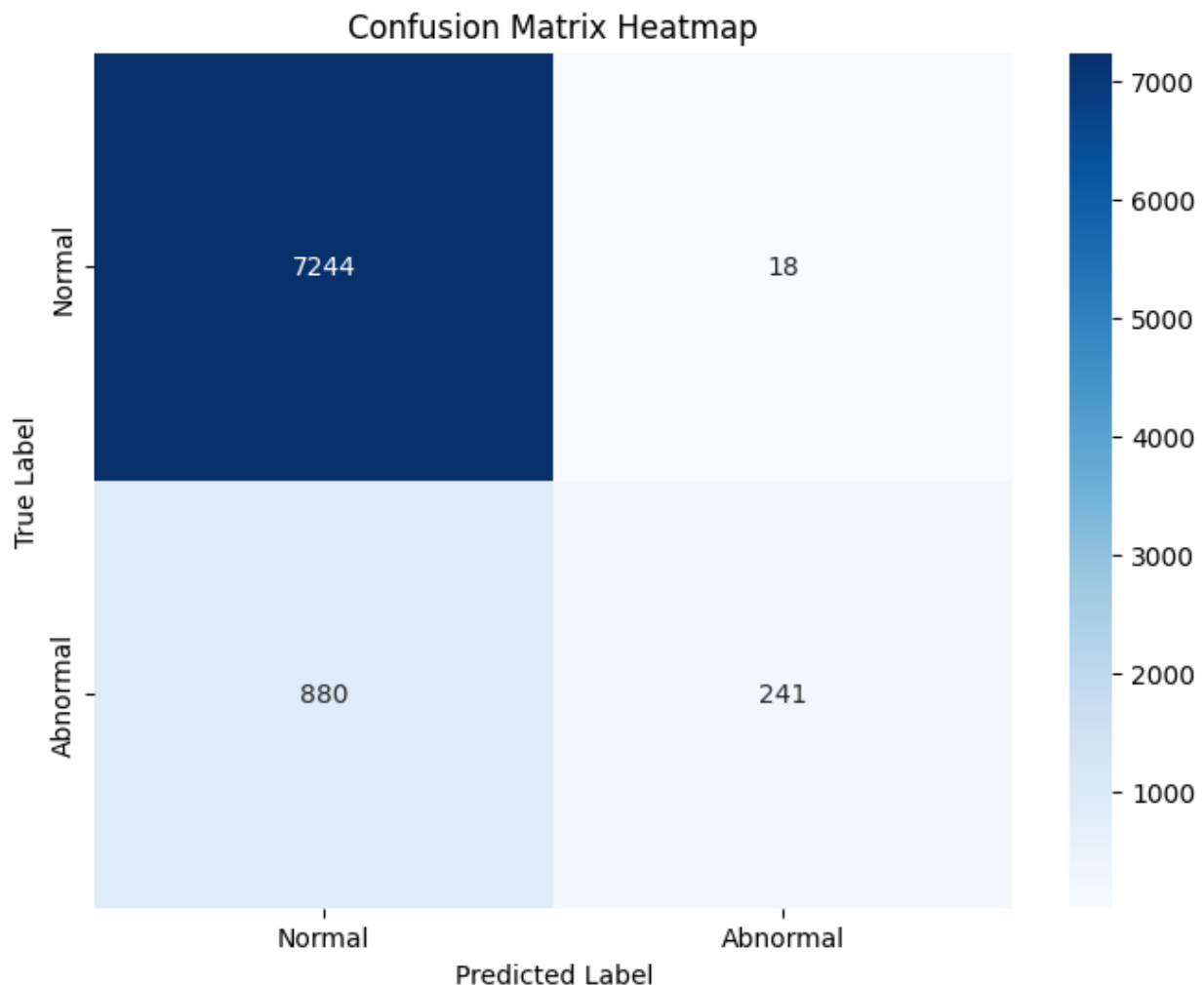
Initial evaluation of the baseline CNN demonstrates that acceptable overall accuracy can be achieved despite poor performance on the minority cancer class. Specifically, while the baseline CNN achieved an overall accuracy of approximately 0.87, it failed to detect any cancer cases, resulting in zero recall and F1-score for the cancer class. This behaviour is reflected in the confusion matrix, where all cancer samples (over 1,100 cases) were misclassified as normal, producing no true positive predictions and a large number of false negatives. This confirms concerns raised during the design phase regarding the inadequacy of accuracy as a sole performance metric. The confusion matrix for the baseline CNN clearly illustrates that the model fails to reliably detect cancer cases.

Figure 18.BASIC CNN CONFUSION MATRIX



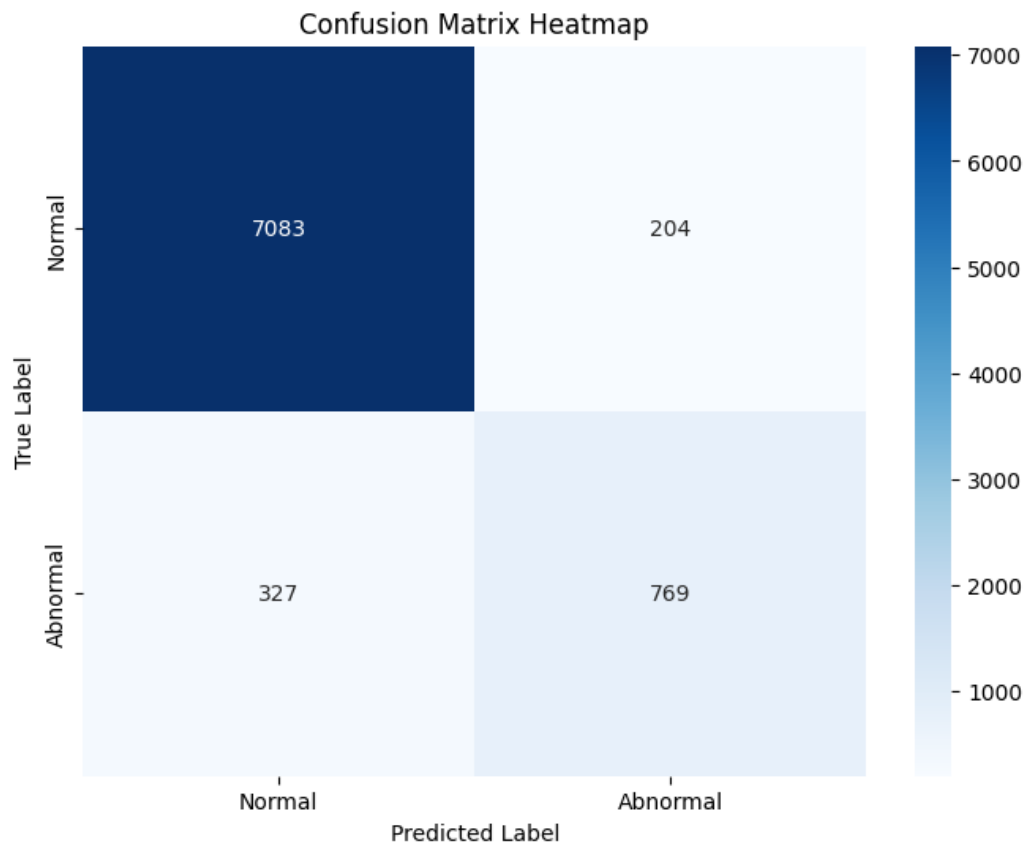
Enhancing the CNN architecture improves feature extraction capacity and leads to moderate gains in performance; however, class imbalance continues to limit recall for the cancer class. The enhanced CNN improves cancer recall to approximately 0.21, corresponding to roughly 240 correctly identified cancer cases, while still misclassifying a substantial number of cancer samples as normal. The ROC-AUC also increases to approximately 0.93, demonstrating better discriminative ability than the baseline model, but still leaving a large proportion of cancer cases undetected.

Figure 19. ENHANCED CNN CONFUSION MATRIX



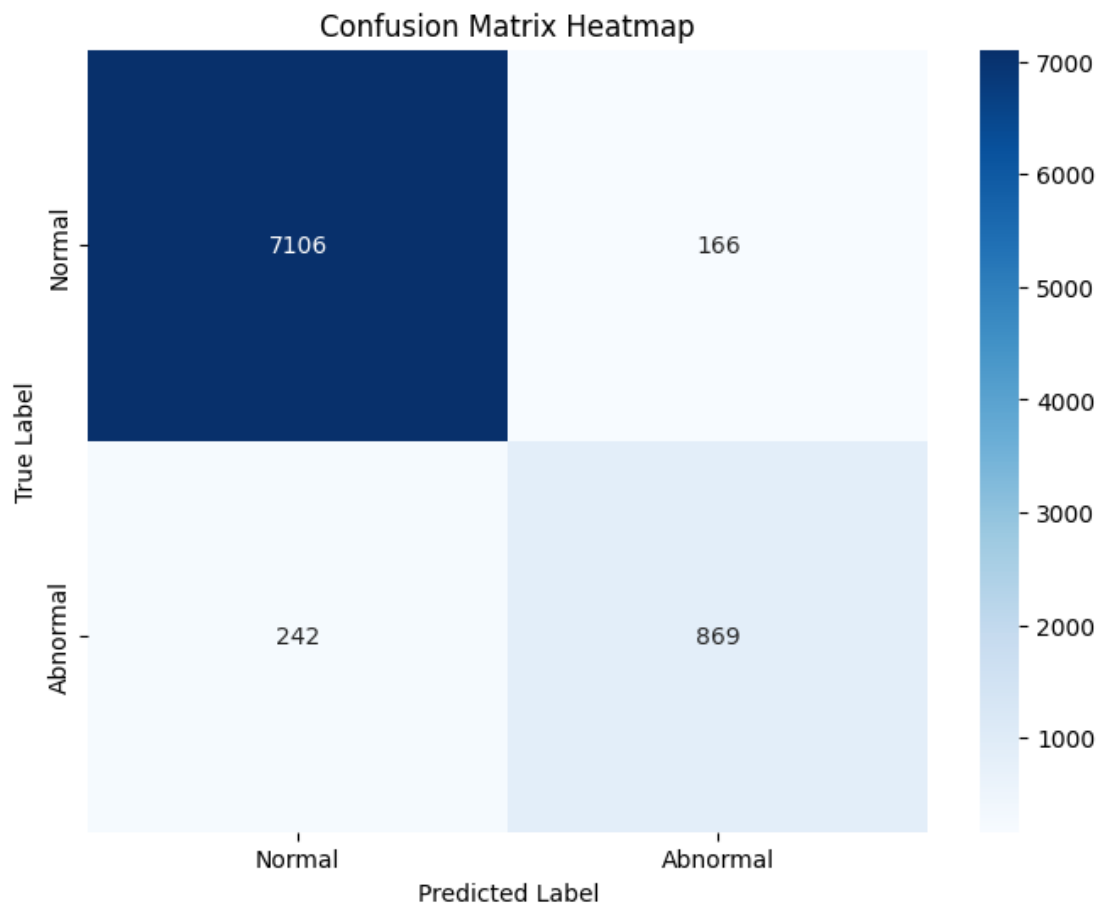
The introduction of transfer learning models, as proposed in the Design section and motivated by findings in the literature, results in substantial improvements in discriminative performance. Compared to the enhanced CNN, transfer learning architectures such as VGG16 and DenseNet121 increase cancer recall to approximately 0.48–0.50, corresponding to correctly identifying around 530–560 cancer cases. These improvements are accompanied by higher ROC–AUC values, indicating stronger class separability. Among the evaluated architectures, the frozen ResNet50 model demonstrates the strongest baseline results, achieving a cancer recall of approximately 0.70, which corresponds to around 770 true positive cancer predictions, and a ROC–AUC close to 0.97. As a result, this model is selected for further optimisation.

Figure 20.FROZEN RESNET50 CONFUSION MATRIX



Applying threshold optimisation to the ResNet50 model leads to a notable shift in performance, particularly in recall and F1-score for the cancer class. By adjusting the decision threshold, cancer recall increases from approximately 0.70 to 0.74, increasing the number of correctly detected cancer cases from around 770 to approximately 785, while maintaining similar overall accuracy. This confirms the design assumption that fixed decision thresholds are suboptimal in imbalanced medical classification tasks. The final optimised model, combining fine-tuning and balanced training, achieves the most favourable trade-off between recall and precision, correctly identifying approximately 870 cancer cases, while reducing false negatives to around 240, and achieving a cancer recall of 0.78, a precision of 0.84, and an F1-score of 0.81.

Figure 21. BEST DDSM MODEL CONFUSION MATRIX



ROC curve analysis further supports these findings by illustrating improved class separability in the final model compared to earlier stages, with the final optimised model achieving a ROC–AUC of approximately 0.98.

Figure 23. ROC OF BASIC CNN

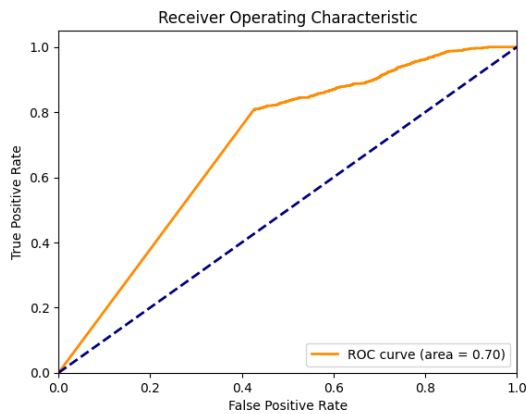


Figure 22. ROC OF ENHANCED CNN

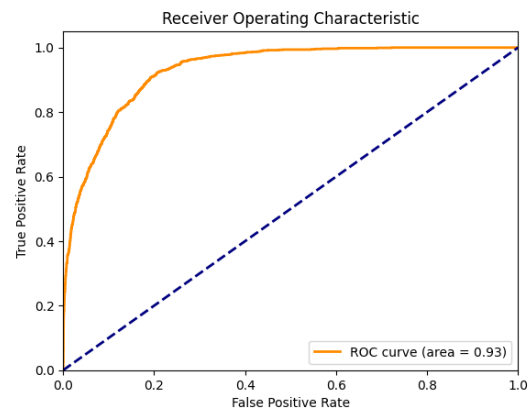


Figure 24. ROC OF FROZEN RESNET50 MODEL

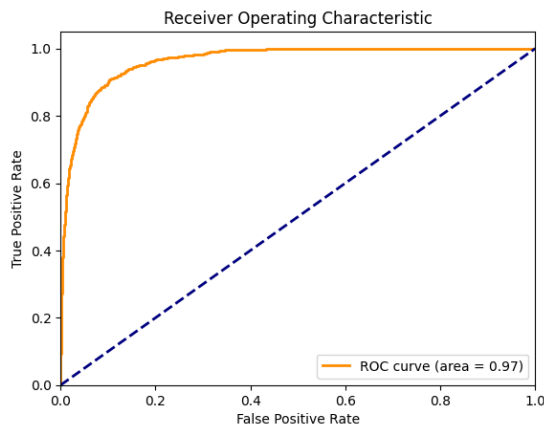
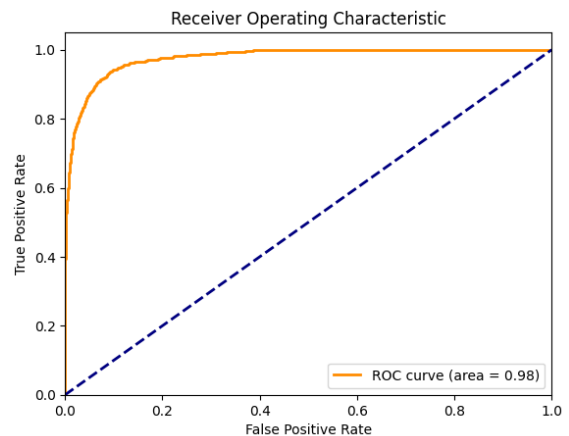


Figure 25. ROC OF FINAL DDSM BEST MODEL



5.3 Impact of Optimisation Techniques on Performance

Beyond architectural selection, several optimisation techniques were implemented to address limitations identified during initial evaluation. These techniques correspond directly to the methods described in Section 4.4 of the Implementation chapter and include threshold optimisation, fine-tuning of pre-trained models, and minority class oversampling.

Table 9 extends the optimisation analysis by reporting full classification metrics at each optimisation stage, highlighting how recall improvements are balanced against precision, accuracy, and overall discriminative performance.

Table 9. Effect of Optimisation Techniques on DDSM Performance

Optimisation Stage	Accuracy	Precision (Cancer)	Recall (Cancer)	F1-score (Cancer)	ROC – AUC	False Negatives
Before optimisation (ResNet50 frozen)	0.94	0.79	0.70	0.74	0.97	327

After threshold optimisation	0.94	0.77	0.74	0.75	0.97	281
After fine-tuning	0.94	0.78	0.75	0.76	0.97	235
After minority class oversampling (Final)	0.95	0.84	0.78	0.81	0.98	242

Threshold optimisation plays a critical role in reducing false negatives by adjusting the decision boundary to align with the project's medical objective. When applied to the frozen ResNet50 model, threshold optimisation alone reduces the number of false negatives from approximately 327 to 281, while increasing cancer recall from 0.70 to 0.74. Fine-tuning further improves performance by adapting pre-trained features to the specific characteristics of mammography images, reducing false negatives to approximately 235 and increasing recall to around 0.75.

Minority class oversampling mitigates training bias caused by class imbalance by increasing exposure to cancer samples during learning. When combined with fine-tuning and threshold optimisation, oversampling results in the final model achieving the highest cancer recall (0.78) and the most balanced precision–recall trade-off, despite a small increase in false positives.

The cumulative improvement achieved through these techniques validates the optimisation strategy proposed during system design and demonstrates

that meaningful performance gains can be achieved without altering the core architecture alone

5.4 Dataset 2: Performance Under Improved Class Balance

As outlined in the Design and Implementation sections, a second dataset was introduced to examine how model performance changes under improved class balance and alternative preprocessing strategies. This dataset is not intended to replace the DDSM benchmark but to provide insight into achievable performance under more favourable data conditions.

Table 10 summarises the classification performance achieved on the second dataset, which features a more balanced class distribution and different preprocessing characteristics compared to DDSM.

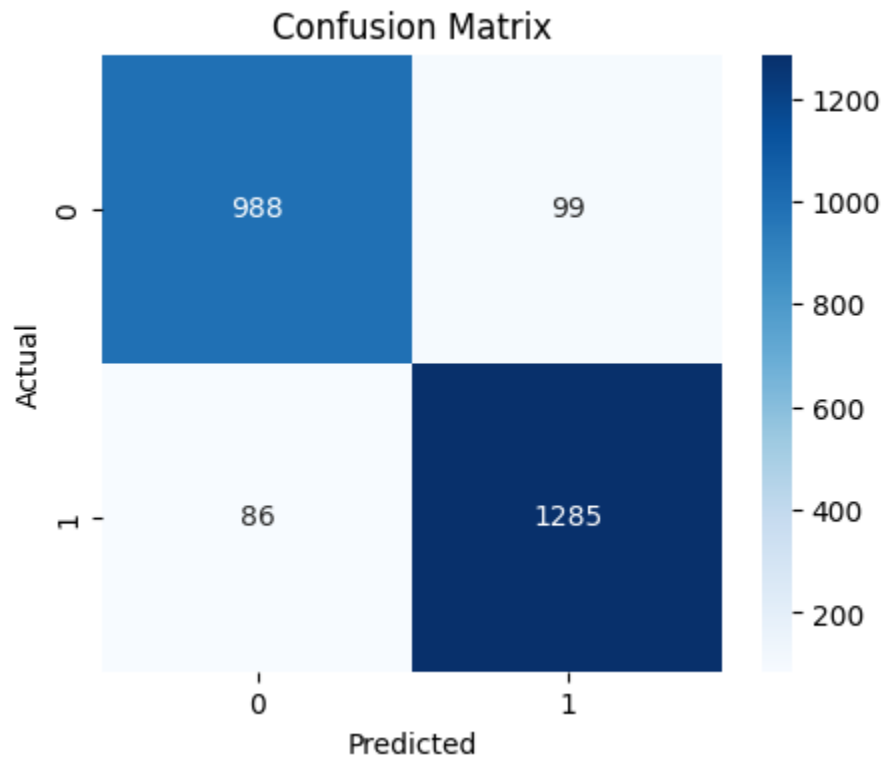
Table 10. RESNET50V2 (DATASET 2) MODEL PERFORMANCE

Metric	Value
Accuracy	0.92
Precision (Malignant)	0.93
Recall (Malignant)	0.94
F1-score (Malignant)	0.93
ROC-AUC	0.981

Performance on this dataset is consistently higher across all evaluated metrics. This improvement is expected and can be attributed primarily to the

more balanced class distribution and the application of data augmentation. A confusion matrix or ROC curve further illustrates the strong classification performance achieved under these conditions.

Figure 26.CONFUSION MATRIX OF RESNETV2 MODEL



These results support conclusions drawn in the literature review regarding the strong influence of dataset balance on reported performance and highlight the importance of contextualising evaluation outcomes.

5.5 User Interface (UI) Evaluation

In addition to quantitative model evaluation, the deployed user interface is evaluated to assess system usability and end-to-end functionality. As described in the Implementation section, the deployment component was designed to demonstrate practical interaction rather than clinical readiness.

An informal usability evaluation was conducted with a small group of non-expert users, including friends and family members. Participants with no prior experience in machine learning systems were invited to interact with the deployed Streamlit application. Each participant was asked to upload a mammography image and observe the classification result generated by the model. After interacting with the system, participants were asked to provide feedback on three specific aspects of the interface: ease of navigation, clarity of the prediction output, and overall usability of the application. Feedback was collected through short surveys immediately after the interaction. This process allowed the evaluation to examine whether the system could be understood and used by non-technical users, which is important for demonstrating the practical usability of the deployed prototype.

Figure 27.STREAMLIT APP UI

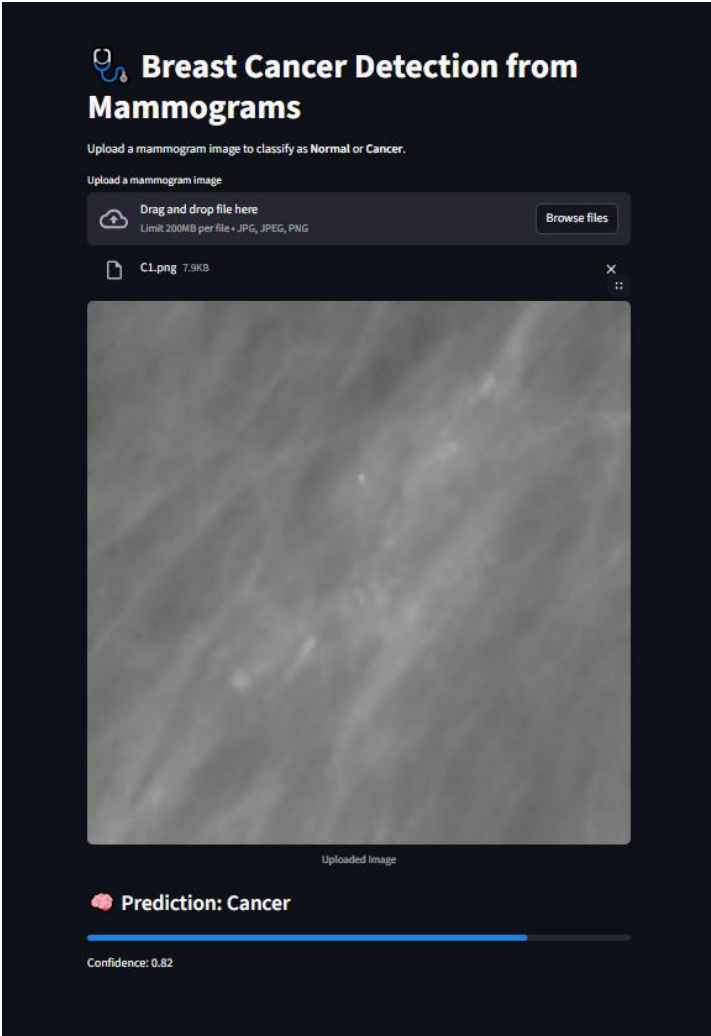


Table 11 summarizes qualitative feedback obtained from the informal usability assessment of the deployed user interface. The questionnaire results are illustrated in **Figures 28–30**

Table 11. Summary of Informal User Interface Feedback

Aspect Evaluated	Feedback Summary
Ease of use	9/10 participants reported the interface was easy or very easy to navigate
Clarity of prediction output	10/10 participants reported that prediction results were clear or very clear
Overall usability	9/10 participants rated the application as excellent or good

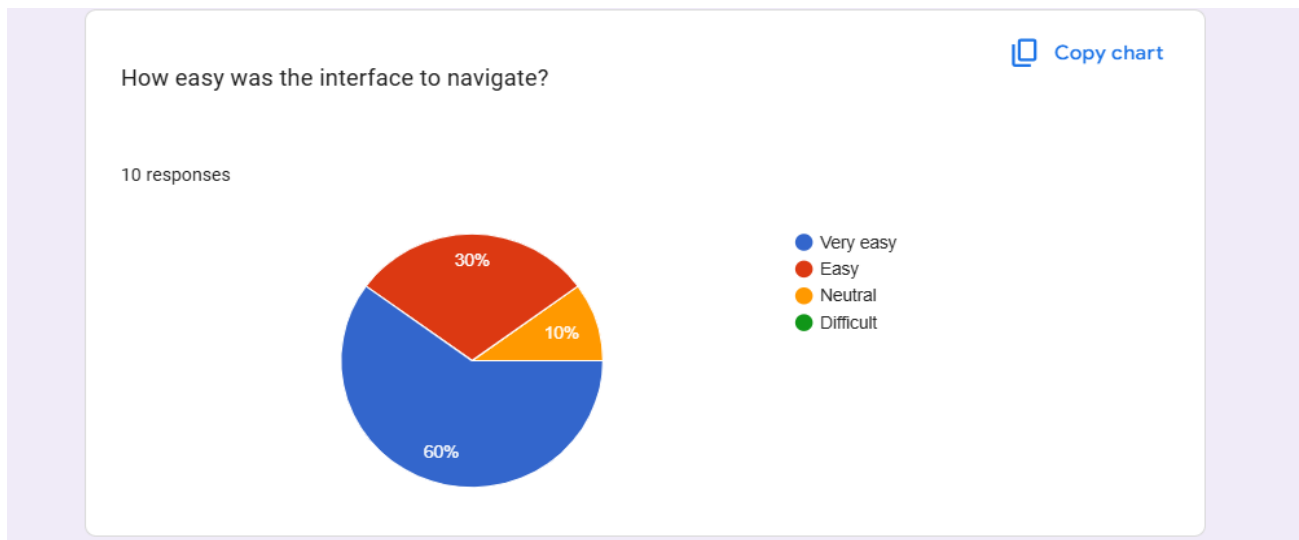
Figure 28. Participant responses to the question evaluating the overall navigation of the deployed Streamlit application

Figure 29. Participant responses evaluating the clarity and interpretability of the prediction output produced by the deployed application

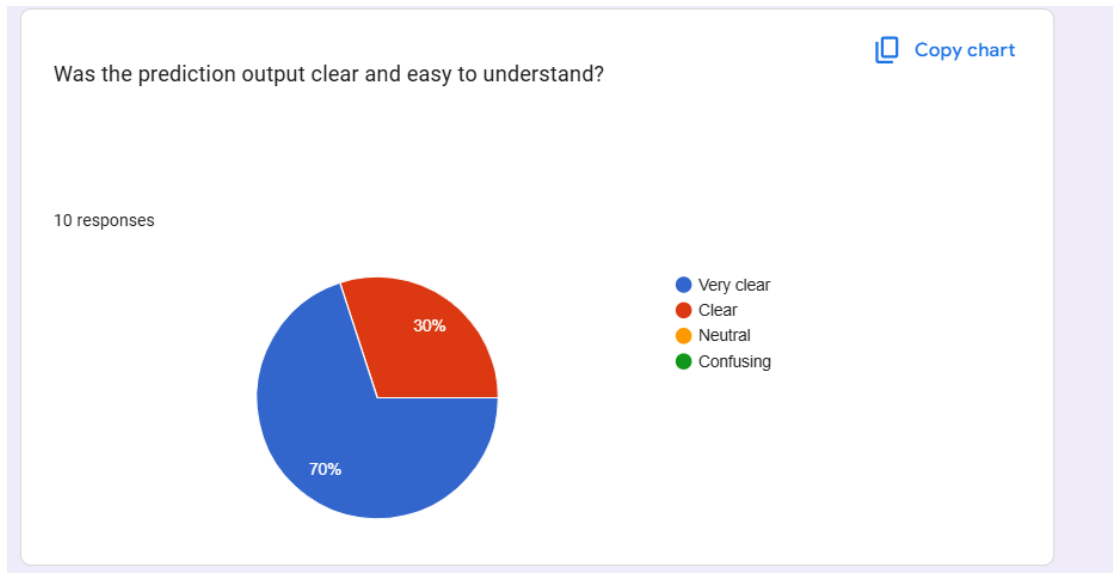
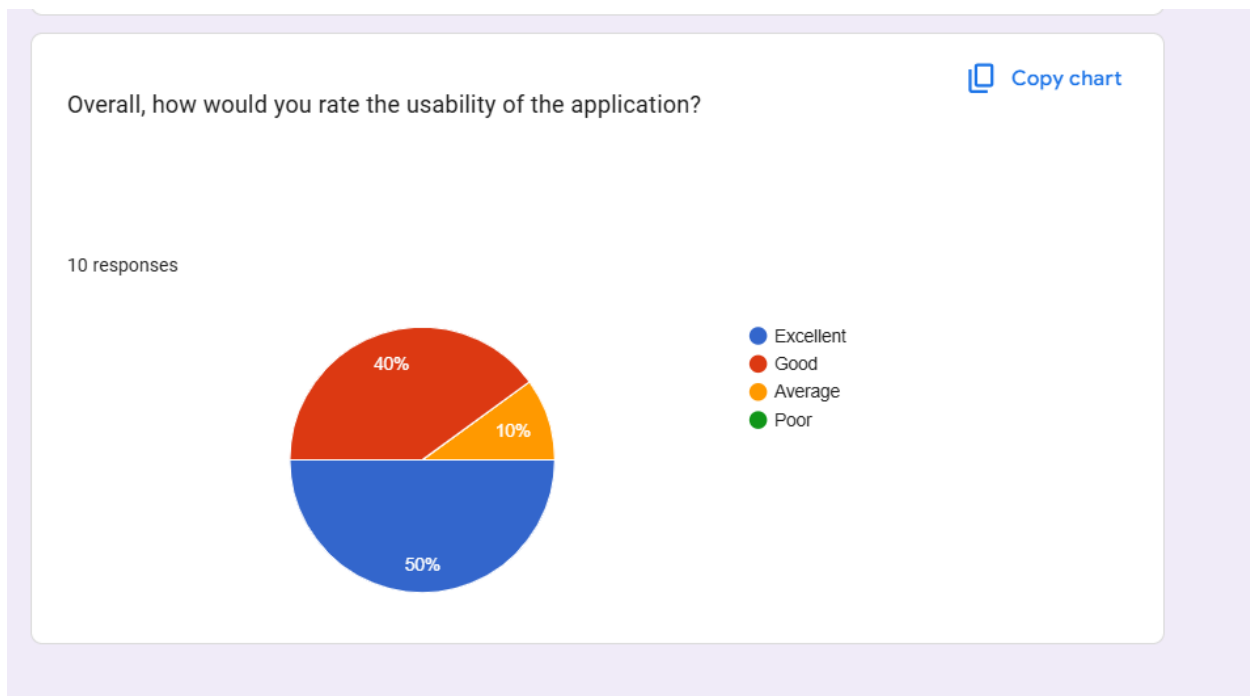


Figure 30. Participant responses to the usability question evaluating the overall usability of the deployed Streamlit application



This evaluation confirms that the system functions as intended from input to output. However, this feedback does not constitute a formal usability or clinical evaluation and is included solely to demonstrate deployment feasibility.

5.6 Cross-Dataset Comparison

To synthesise findings across datasets, a direct comparison is made between the final optimised DDSM model and the model trained on the second dataset.

Table 12 provides a direct comparison between the final optimised DDSM model and the model trained on the second dataset, using consistent evaluation metrics across both datasets.

Table 12. CROSS DATASET MODEL COMPARISON

Metric	Final DDSM Model (Fine-tuned ResNet50 + Balanced)	Dataset 2 Model (ResNet50V2)
Accuracy	0.95	0.92
Precision (Positive Class)	0.84	0.93

Recall (Positive Class)	0.78	0.94
F1-score	0.81	0.93
ROC-AUC	0.98	0.981

The model trained using the second dataset yields better overall performance due to balanced data and augmentation. The recall and f1-score of the final DDSM-based model for the malignant class were 0.78 and 0.81, respectively, as opposed to about 0.94 and 0.93 for the second model and its performance. The ROC-AUC for Dataset 2 is around 0.981, which is only marginally higher than the 0.98 score of the DDSM final model.

The final DDSM model is fairly resistant to high class imbalance, as a relatively small proportion of the dataset is cancer. This is seen in the outputs, where under high class imbalance settings the cancer recall is at 0.78 and precision is at 0.84. This suggests that the model is able to handle realistic cancer distributions.

5.7 Critical analysis

The evaluation results demonstrate that the proposed project design and optimisation pipeline successfully improved breast cancer classification performance across multiple stages. The progressive modelling strategy proved particularly effective. Beginning with a baseline CNN provided a clear reference point for evaluating improvements, while the introduction of

transfer learning architectures produced the most substantial gains in cancer detection. The baseline CNN achieved an overall accuracy of approximately 0.87 but failed to detect any cancer cases, resulting in zero recall for the cancer class. In contrast, transfer learning models such as ResNet50 significantly improved performance, increasing cancer recall to approximately 0.70 while achieving ROC–AUC values close to 0.97. This confirms that pre-trained deep convolutional networks provide more effective feature extraction for medical imaging tasks than shallow CNN architectures trained from scratch.

Additional optimisation techniques further enhanced model performance. Threshold optimisation improved recall from approximately 0.70 to 0.74 by adjusting the classification decision boundary, while fine-tuning allowed the model to adapt high-level features to mammography-specific patterns. When combined with minority class oversampling, the final fine-tuned ResNet50 model achieved a cancer recall of approximately 0.78 and a ROC–AUC of around 0.98, demonstrating that the optimisation pipeline effectively improved sensitivity to malignant cases despite severe class imbalance.

The use of two datasets also provided valuable insights into the impact of dataset characteristics on model performance. While the DDSM dataset served as a realistic benchmark due to its severe class imbalance, the second dataset demonstrated that substantially higher performance metrics can be achieved under more balanced data conditions. This comparison highlights the strong influence of dataset composition on reported model performance and reinforces the importance of evaluating medical imaging systems under different data conditions.

Despite these strengths, several limitations remain. Severe class imbalance continues to constrain achievable recall on the DDSM dataset, even after applying oversampling and threshold optimisation. Computational constraints limited the scope of hyperparameter exploration and model experimentation. In addition, differences in dataset acquisition protocols and preprocessing methods may affect cross-dataset generalisation. Finally, the

system was evaluated using retrospective datasets rather than prospective clinical studies, meaning that real-world diagnostic performance cannot be fully assessed.

Overall, the project demonstrates that a carefully designed deep learning pipeline combining transfer learning, threshold optimisation, fine-tuning, and class balancing can significantly improve breast cancer detection performance while highlighting ongoing challenges in medical image classification.

Table 13. Summary of Project Strengths and Limitations

Strengths of the Project	Limitations and Areas for Improvement
Progressive optimisation pipeline improved model performance across multiple stages	Severe class imbalance in the DDSM dataset continues to affect cancer recall
Transfer learning models significantly improved discriminative performance	Hyperparameter tuning and fine-tuning were limited by available computational resources
Final model achieved strong ROC–AUC performance (~0.98)	Generalisation across datasets is limited due to differences in acquisition and preprocessing
Threshold optimisation and oversampling improved cancer detection performance	Evaluation was conducted on retrospective datasets rather than real-world clinical data
Deployment through a Streamlit interface demonstrated end-to-end system usability	User interface evaluation was informal and involved a small number of participants

5.8 Summary of Findings and Implications

The results of the evaluation showed that the proposed optimisation pipeline improved the classification performance on breast cancer diagnosis. The results also highlighted the trend from baseline CNN architectures to transfer learning, threshold optimisation, fine tuning and oversampling the minority classes. The final fine-tuned ResNet50 model achieved strong discriminative performance on the DDSM dataset (ROC-AUC score of around 0.98), with reasonable cancer recall rates despite the extreme dataset imbalance.

The results also show the importance of using recall-focused evaluation metrics in medical classification tasks where false negatives must be avoided, rather than regular performance metrics such as accuracy. This is further shown with the second dataset where class balancing and data augmentation methods can influence metrics reported.

Ultimately, the project has successfully created a strong classification pipeline to classify malignant mammography images while appropriately dealing with class imbalance. It provides groundwork for further work, such as validating the pipeline on other datasets and further refining hyper-parameters for the model within the pipeline.

6. Conclusion

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This research investigated the use of deep learning methods for breast cancer detection in mammograms, in order to improve classification accuracy in class imbalance scenarios, and reduce the possible generation of false negatives. The experimental pipeline was designed as a combination of baseline architectures through to advanced architectures, transfer learning, threshold optimisation, fine-tuning and data balancing.

Due to its extreme class imbalance and clinical focus, the DDSM dataset was used as the main benchmark throughout this project. Experiments also found that cancer detection performance does not always correlate with the accuracy. As a result, accuracy should not be the only metric assessed when measuring the performance of medical screening tasks. This led to prioritizing recall-based evaluation metrics and confusion matrix analysis in the decision-making process, impacting design and optimization.

Transfer learning provided a substantial improvement to the performance of the models, with ResNet50 being the most effective of the architectures assessed. The performance of the models was additionally improved through threshold adjustment, model fine tuning, and oversampling of the minority class, with the latter reducing false negative predictions. The best balanced, fine-tuned ResNet50 model was the top performing DDSM dataset model, hence, it was chosen to be deployed.

A second set of breast mammography images with masses was made available to investigate if class balancing or alternate pre-processing would increase the robustness of the model. While the performance improved numerically, it was not due to increased robustness but rather caused by the

class balancing and data augmentation itself. Per above project template and goals, DDSM is considered primary. Finally, the best classifier identified in the DDSM pipeline was reported here. Since the ultimate goal is to consider clinical applicability in real screening conditions, a strong classifier is required where abnormalities are rare and false negatives costly.

The project also demonstrated end-to-end system functionality through the use of the final model in an interactive application. Note that this application is not clinically validated, and is only included for demonstration purposes to show how machine learning models can be deployed within a usable system.

The project successfully developed a reliable and strong deep learning pipeline for breast cancer detection and confirmed findings from the wider literature on the challenges of class imbalance, the limitations of accuracy as an evaluation metric and the need for model optimisation strategies to be aligned with domain specialists.

Possible areas of further research could include use of larger, multi-institution datasets, alternative strategies for dealing with the difficulties of class imbalance, more advanced neural network explainability methods, and additional clinical validation under specialist supervision before translation into clinical use. Nevertheless, through this work, a framework has been provided for more work in this area, along with the practical challenges in applying deep learning to medical image analysis.

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